

CHAPTER 14

ADMINISTRATION, SCORING, AND INTERPRETATION OF SELECTED TESTS

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The authors would like to acknowledge the significant contributions of Michael McGuigan, Everett Harman, and John Garhammer to this chapter.

After completing this chapter, you will be able to

- discern the best ways to measure selected parameters related to athletic performance;
- administer field tests appropriately;
- evaluate and analyze test data and make normative comparisons;
- understand appropriate statistics; and
- combine the results of selected tests to generate an athletic profile.

As discussed in [chapter 13](#), the strength and conditioning professional—often referred to as the *tester* in this chapter—who has a broad understanding of exercise science can effectively choose and use tests and measurements to make training program decisions that help athletes optimize their physical preparation and maximize their potential. To do this effectively, the tester must administer tests correctly, analyze test data accurately (or understand how test data have been analyzed if the tester is not responsible for this, such as when using proprietary software), and then combine the results of selected tests to generate an athletic profile. This chapter covers these basic aspects of testing performance-related parameters and provides comprehensive age- and sport-specific descriptive and normative data for selected tests.

Measuring Parameters of Athletic Performance

Athleticism incorporates many physical abilities, some of which are much more amenable to training than others. Such abilities may be called general components of **athletic performance**—that is, abilities that enable effective responses to the various physical demands of a specific sport or event. This section focuses on how each component can be tested and highlights relevant issues.

Maximum Muscular Strength (Low-Speed Strength)

Maximum muscular strength tests can be split into dynamic or isometric demands; those that involve relatively low movement speeds reflect **low-speed muscular strength**. In this case, muscular strength is related to the force a muscle or muscle group can exert in one maximal effort while maintaining proper form, and it can be quantified by the maximum weight that can be lifted once (the 1-repetition maximum [1RM]) in exercises such as the bench press or back squat, or the maximum force that can be exerted at a particular isokinetic velocity (5, 6, 45, 72, 100, 101, 103, 109, 124). Because 1RM tests do not require expensive equipment and reflect the kind of dynamic ability necessary in sport, they are the maximal strength tests of choice for most strength and conditioning professionals.

In general, 1RM tests are administered after the athlete has warmed up by performing a few sets of the test exercise with submaximal loads, beginning with a relatively light load. The first attempt is usually with approximately 50% of the athlete's estimated 1RM weight. After the athlete has rested enough to feel recovered from the previous attempt (1-5 min, depending on the difficulty of the attempt), the strength and conditioning professional increases the weight somewhat, based on the ease with which the previous trial was performed. A skilled strength and conditioning professional should, within three to five attempts following warm-up, be able to find the athlete's 1RM load to within a few percentage points of the true value.

While testing the athlete's 1RM is a time-tested and reliable way to assess maximal strength, many strength and conditioning professionals have begun to use isometric tests to more comprehensively assess the athlete's maximal force production capacities (5, 6, 19, 29, 45, 72, 100, 101, 103, 124). These types of tests are particularly appealing because they are nonfatiguing and are often integrated into an athlete's regular training program (i.e., training is testing). Typically administered with the use of a force plate (or pair of force plates) or force cell that is integrated into a custom testing device, these tests allow for the generation of a force-time curve that can be used to quantify the rate at which force is produced as well as the maximum force exerted against an immovable object (e.g., bar) (29, 92). Regardless of the isometric testing device used, these tests are performed after a suitable warm-up, often progressing from 50% effort to 90% before performing three maximum isometric trials that are performed for 3- to 5-second duration (29, 59).

Given that strength is highly influenced by body mass, expressing maximum muscular strength test results relative to body weight is valuable. Ratio standards can be used to determine relative strength (43) by dividing the weight lifted (in a dynamic test) or force produced (in an isometric test) by the athlete's body weight. Relative strength from a 1RM test can be expressed as the total weight lifted (lb or kg) divided by body mass (lb or kg). To express the same from an isometric test, maximal force exerted in newtons is divided by bodyweight converted into newtons ($\text{lb} \times 4.44$, or $\text{kg} \times 9.81$). Ratio scaling allows comparisons between athletes normalized to body weight. However, this simple approach assumes that strength is directly proportional to body mass (43), which is not necessarily the case. Therefore, allometric scaling has also been recommended, particularly in settings with wide variation in body size and strength. Dividing strength by body mass raised to an appropriate power can eliminate the effects of body mass, although debate remains regarding the most appropriate scaling exponents (33, 43, 83).

Anaerobic or Maximum Muscular Power (High-Speed Strength)

High-speed muscular strength or **maximal anaerobic muscular power** (or **anaerobic power**) is related to the ability of muscle tissue to exert high force while contracting at a high speed. Tests of strength and power are of very short duration, are performed at maximal movement speeds, and produce very high-power outputs. High-speed maximal muscular power tests are often called (*maximal*) *anaerobic power tests*. Scores on high-speed muscular strength tests include the 1RM of explosive exercises (e.g., the power clean, snatch, push jerk), the height of a vertical jump, and the time to sprint up a staircase (e.g., Margaria-Kalamen test) (69, 100, 103, 124, 126). Because explosive exercise tests take about 1 second while low-speed maximal strength tests generally require 2 to 5 seconds to complete, phosphocreatine and adenosine triphosphate (ATP) stored in the active muscle(s) are the primary energy sources for both types of tests. Maintaining correct technique or form is also important when measuring anaerobic power for both performance validity and safety reasons.

Most maximal muscular strength tests involve relatively slow movement speeds and therefore reflect low-speed strength. Assessment of high-speed muscular strength can involve measuring the 1RM of explosive resistance training exercises or the height of a vertical jump.

Power output reflects the interplay between force and velocity. The height of a jump is a function of the force applied to the ground that is greater than the athlete's body weight over the time it was applied (this is referred to as *net impulse*) while he or she extends the ankles, knees, and hips to propel upward. A larger net impulse (force $\times$ time) leads to a higher velocity at which the athlete leaves the ground. An athlete may not improve in jump height after gaining body weight during a resistance training cycle, making it appear that power output is unchanged. However, because the athlete is heavier and propels the body to the same height, indicating the same takeoff velocity, an increase in power output is likely. This applies to any test in which body weight is manipulated (e.g., running up stairs). Moving a heavier body at the same speed requires a higher power output. It is up to the strength and conditioning professional to interpret the interplay between changes in body weight, velocity, and power output accordingly, because for some athletes, such as those from collision-based sports, gains in absolute power through weight gain may be beneficial, but for athletes who benefit primarily from moving at high absolute velocities, a reduction in velocity owing to weight gain may be detrimental.

An alternative class of anaerobic power tests involves the use of a cycle ergometer. This type of test can be advantageous for the strength and conditioning professional in some injury situations in which running is restricted or when the athlete participates in a non-bodyweight-support sport such as rowing or cycling. The most commonly used test of this type is the Wingate anaerobic test. A field test protocol involves use of a cycle ergometer with mechanical means of adjusting resistance and measuring pedal revolutions (i.e., revolutions per second). In a laboratory setting, an electronically instrumented ergometer can simplify parameter measurement and improve accuracy. Typical protocols involve a basic warm-up followed by a 30-second test interval (39). In this test, resistance is applied quickly after the individual reaches a near-maximal pedaling rate (typically 90-110 rpm). The resistance applied is proportional to body weight (e.g., 7.5%); the percentage is greater for trained athletes than for individuals with less training. Work performed is determined from the resistance value and number of pedal revolutions. Power is generally calculated as work divided by time for each 5-second time interval during the 30-second test. Parameters typically

calculated include peak power, average power, and a fatigue index such as a ratio of maximum to minimum interval power. Norms for cycle ergometer tests are available (71, 103).

Load-Velocity Profile

Velocity-based training (VBT) using commercially available technology has become an increasingly popular method for measuring bar speed and displacement during the performance of individual resistance training exercises. These technologies allow testers to measure the intensity of an exercise using bar velocity as a metric rather than simply using percentages of a 1RM (15, 156). These methods are based on the inverse relationship between bar speed and the weight lifted. Specifically, as the absolute weight on the bar increases, a linear decrease in bar velocity is typically recorded. Using this knowledge, strength and conditioning professionals can create an individualized load-velocity profile by performing a series of sets performed with progressively increasing loads (i.e., 20%, 40%, 60%, 80%, and 90% of 1RM) (156). The creation of the load-velocity profile has been suggested to be useful for evaluating and prescribing exercise intensity during resistance training programs and has also been suggested to be a useful tool for estimating or monitoring the 1RM. When the load-velocity profile is created from a series of increasing loads as a percentage of the 1RM with the back squat exercise ([figure 14.1](#)) a linear relationship exists between load and velocity (14). However, when the 1RM back squat is estimated from the load-velocity profile, it is highly variable and consistently results in an overestimation of the athlete's 1RM and should not be used for 1RM prediction (56, 89). When the current body of scientific knowledge is examined using the load-velocity profile to predict the 1RM on a day-to-day basis, it may result in the misprogramming of training loads and, in some instances, the programming of loads that exceed the athlete's capacities (14, 56, 60). As such, it is generally recommended that the 1RM should be directly tested when seeking to determine an athlete's maximal strength in a given exercise, and the load-velocity profile should not be used to estimate the 1RM (56, 89).

Anaerobic Capacity

Anaerobic capacity is the maximal rate of energy production by the combined phosphagen and anaerobic glycolytic energy systems for moderate-duration activities. It is typically quantified as the maximal power output during muscular activity between 30 and 90 seconds using a variety of tests for the upper and lower body (39, 103, 124, 151), as opposed to maximal anaerobic power tests, which last no longer than a few seconds.

Local Muscular Endurance

Local muscular endurance is the ability of certain muscles or muscle groups to perform repeated contractions against a submaximal resistance (18, 103). A test of local muscular endurance should be performed in a continuous manner for several seconds to several minutes without the advantage of rest periods and without extraneous body movements. Examples include performing a maximal number of repetitions of the chin-up, parallel bar dip, or push-up exercises or a resistance training exercise using a fixed load (e.g., a percentage of an athlete's 1RM or body weight) (38, 46, 94, 100, 103).

Aerobic Capacity

Aerobic capacity, also called **aerobic power**, is the maximum rate at which an athlete can produce energy through oxidation of energy sources (carbohydrates, fats, and proteins) and is usually expressed as a volume of oxygen consumed per kilogram of body weight per minute (i.e., $\text{mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$) (95). Few strength and conditioning professionals have the equipment to measure oxygen consumption directly, so aerobic capacity is generally estimated by performance in aerobic endurance activities such as running 1 mile (1.6 km) or more (69, 111, 120). It can also be

estimated using other field tests such as the maximal aerobic speed (MAS) test (87) and the Yo-Yo intermittent recovery test (10, 18, 85, 86).

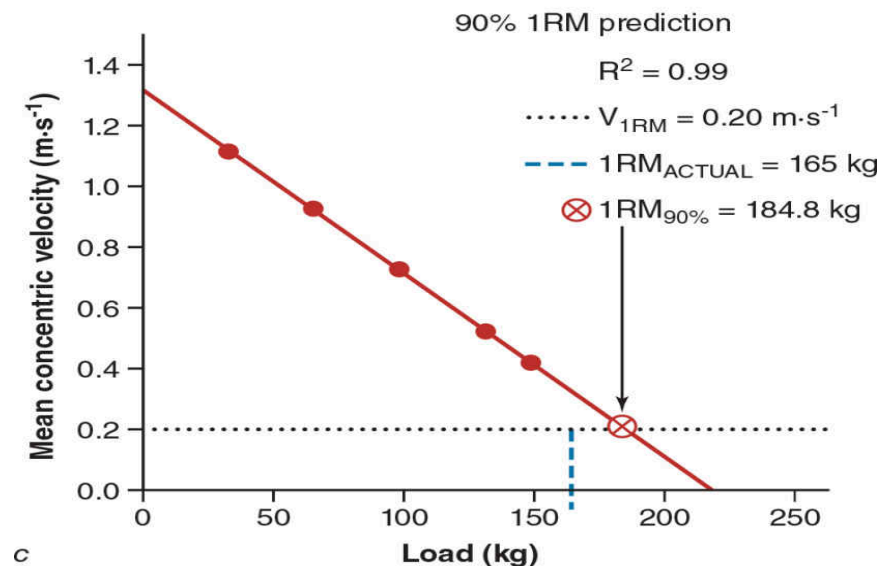
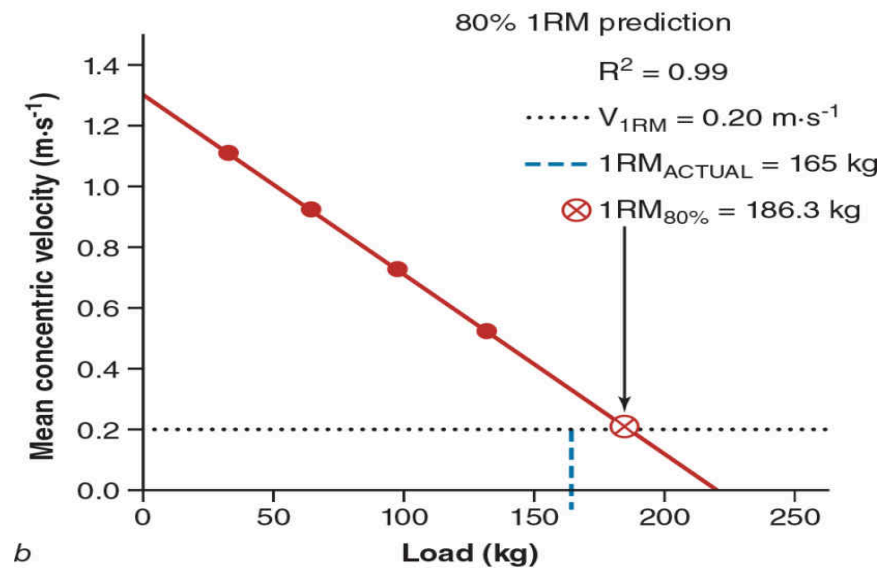
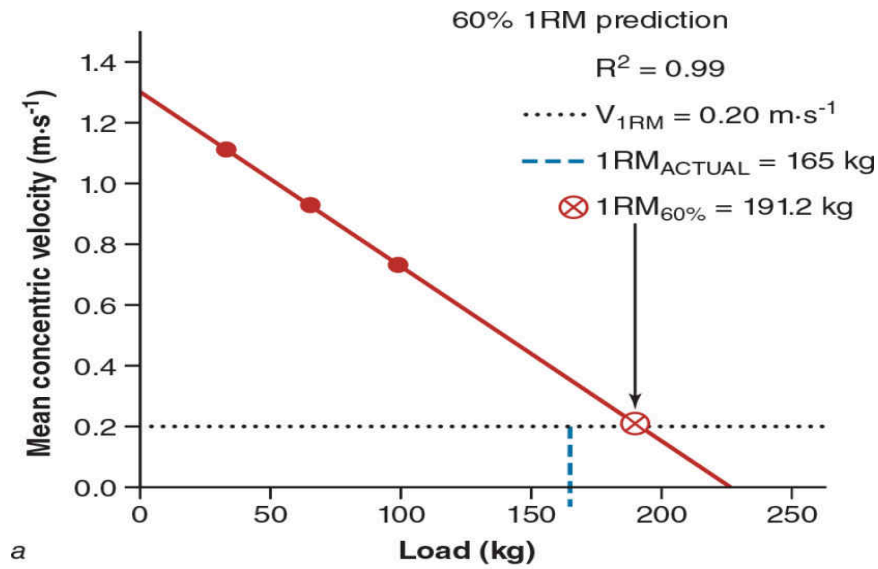


FIGURE 14.1 Load–velocity profiling.

Reprinted by permission from H.G. Banyard, K. Nosaka, and G.G. Haff, "Reliability and Validity of the Load-Velocity Relationship to Predict the 1RM Back Squat," *The Journal of Strength and Conditioning Research* 31, no. 7 (2017): 1897-1904.

Agility

Agility traditionally has been considered the ability to stop, start, and change the direction of the whole body rapidly (134, 143). Agility consists of two main components: speed in changing direction and cognitive factors (134). The definition of agility has been revised to take into account the perceptual qualities, and it is now considered "a rapid, whole-body, change of direction or speed in response to a sports-specific stimulus" (134, 143). Agility testing therefore involves tests of change of direction with a cognitive component, such as anticipation and reaction. Tests such as the T-test, the 5-0-5 test, the 5-10-5 test (also known as the *pro agility test*), and the L-drill (also known as the *3-cone drill*) are used to assess change of direction.

Speed

Speed is movement distance per unit of time and is typically quantified as the time taken to cover a fixed distance. The time taken to sprint from a stationary start over a short distance such as 10 yards or meters reflects acceleration, whereas longer sprints such as 40 yards (37 m) would measure maximum speed (163). Tests of speed are not usually conducted over distances greater than 100 meters (109 yd) because longer distances reflect anaerobic or aerobic capacity more than absolute ability to move the body at maximal speed (103, 124, 163).

Electronic timing devices are becoming more accessible to strength and conditioning professionals due to increased ease of use and lower prices. However, many tests of speed and agility are administered using hand timing with a stopwatch, which can be a major source of measurement error, especially if the tester is not sufficiently trained. Even under ideal conditions, stopwatch-measured sprint times are up to 0.24 seconds faster than electronically measured times because of the tester's reaction-time delay in pressing the stopwatch button at the gun and the tendency to anticipate and press the button early as the athlete approaches the finish line (45, 67, 125). Therefore, strength and conditioning professionals are encouraged to use electronic timing devices for tests of speed and agility when they are available. It is also more informative to measure split times, because this can provide the strength and conditioning professional with insight into the speed and acceleration capacities of the athletes. For example, times for 10 yards (9 m), 20 yards (18 m), and 40 yards (37 m) can be recorded and used to calculate split times and maximal velocity. Finally, tests of speed and agility require proper footwear and a nonslip surface.

Athletic Motor Skill Competency

Developing movement skills that enable children and adolescents to move in an effective and efficient manner during physical activity and sport is central to long-term athletic development. The **athletic motor skill competencies** (AMSC), proposed by Lloyd and colleagues (90), presents foundational movement skills that underpin athletic movements. The AMSC and examples of associated movement screening exercises are presented in [figure 14.2](#) and can be used to monitor the development of young athletes (123). The use of the AMSC and other screening tools provide a method for tracking athlete development with the use of various assessment tools that evaluate movement patterns and other qualitative aspects of movement. As such, these types of tools are best described as process-oriented assessments (150). Overall, the use of movement screening information can be used by practitioners to identify strength, mobility, and neuromuscular deficits, which can then guide programming decisions.

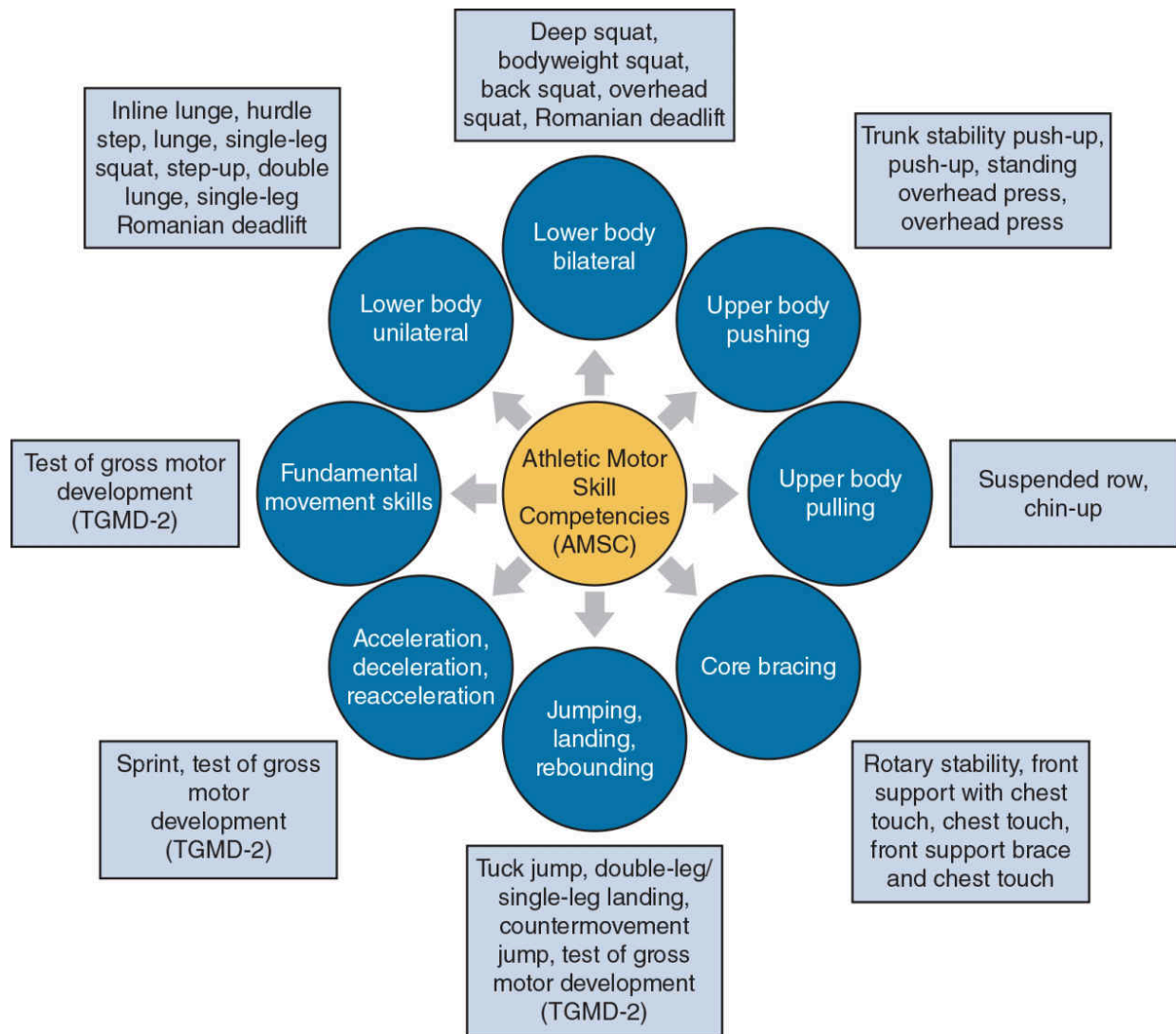


FIGURE 14.2 Athletic motor skill competencies and examples of associated movement screen exercises.

Adapted from Lloyd and Oliver (2020); Pullen et al. (2022).

Assessing movement patterns can reveal indicators of injury susceptibility in young athletes. Well-established screening tools such as the **functional movement screen (FMS)**, tuck jump assessment (TJA), landing error scoring system, and drop jump assessments are commonly used for this purpose (123). Additionally, tailored movement screening protocols have been devised specific to particular sports, aiming to pinpoint injury risks, performance limitations, or even talent identification opportunities. Tailored training programs can be developed by strength and conditioning professionals to target specific individual needs, including addressing movement limitations.

Multiple movement screens assess parts of the AMSC spectrum, but consensus has not been reached regarding which are most appropriate for a given cohort. One narrative review (122) included 10 movement screens, including but not limited to the FMS, TJA, resistance training skills battery (RTSB), and back squat assessment (BSA). Interested readers are directed to the review by Pullen and colleagues (122) for more detailed information on the methodologies, reliabilities, and strengths and limitations of each of these tests when selecting the screening test that is the most appropriate for the environment.

Implementing movement screens can be complex due to variations in implementation practices, criteria, grading, and movement analysis. Understanding reliability of screen implementers is critical, although both novice and expert raters can achieve substantial reliability (104). Recording movements for retrospective analysis after testing is recommended, because it allows raters to review video footage, pause, slow down, and rewatch to accurately assess movements. The number of repetitions performed and how they are graded are crucial. Due to ongoing brain maturation, youth exhibit higher movement variability, and performance regressions can coincide with growth spurts (54, 68). Therefore, movement screens should assess each repetition for consistency. Some screens, such as the FMS and RTSB, use a best repetition technique, which limits the assessment of movement consistency. Others, like the TJA, use a sum-of-repetition scoring system, capturing movement variability. Considering total body versus segmental analysis is also important because total body approaches like the FMS may miss specific deficiencies or compensations, whereas segmental analysis provides detailed insights (123).

Flexibility

Flexibility can be defined as the range of motion about a body joint (13). Typical devices for measuring flexibility include manual and electric goniometers, which measure joint angle, and the sit-and-reach box, which is used to evaluate the combined flexibility of the lower back and hips. Flexibility measurements are more reliable when standardized warm-up and static stretching precede the flexibility assessment. During a flexibility test, the athlete should move slowly into the fully stretched position and hold this position. Ballistic stretching, characterized by bouncing to increase range of motion, cannot be allowed during any flexibility testing (69, 116).

A number of movement competency screens are available for strength and conditioning professionals and can be used to assess overall flexibility, mobility, and general movement competency of athletes. However, there is no current consensus on which screen to use, nor a clearly established link between movement screening results and injury (98, 116). Competent strength and conditioning professionals perform postural and performance screening routinely by viewing the athletes' performance in training. For example, the overhead squat is a common exercise that is used as a part of movement screens because it allows assessment of bilateral mobility of the hips, knees, and ankles along with the shoulder and thoracic spine (3, 25, 126).

Balance and Stability

Balance is the ability to maintain static and dynamic equilibrium or the ability to maintain the body's center of gravity over its base of support (103, 124). **Stability** is a measure of the ability to return to a desired position following a postural disturbance (103). Athletes with poor balance are at a greater risk of lower limb injuries (76, 77, 96). Athletes have also been shown to have greater balance compared to nonathletes (38). Balance testing can be used to assess stability increases with training and in a number of different ways (103). Commonly used tests include timed static standing tests (eyes closed and standing on one or both legs) (21, 101), balance tests using unstable surfaces (101), and tests using specialized balance testing equipment (e.g., NeuroCom, Biodex Balance System) (124). These include a large number of tests that can evaluate different aspects of balance and stability (103). The balance error scoring system (BESS) and star excursion balance test (SEBT) have very good reliability and a substantial body of literature supporting their use (21, 57, 66, 103, 115, 146).

Body Composition

Body composition usually refers to the relative proportions by weight of fat and lean tissue. Although there are sophisticated and expensive devices capable of partitioning the lean component into bone and soft tissue, the body composition procedures typically performed by strength and

conditioning professionals use the basic two-component (fat and lean mass) model. With a trained and competent tester, the skinfold measurement technique is the most valid and reliable ($r = 0.99$) means for assessing body fatness that is generally available to the strength and conditioning professional and is preferable to body circumference methods (95). Other methods available to strength and conditioning professionals include bioimpedance devices and total body plethysmography, although dual X-ray absorptiometry (DXA) is often labeled as a gold standard.

The skinfold method uses calipers that measure the thickness of a double layer of finger-pinch skin and subcutaneous fat. A good skinfold measurement device should squeeze the fold of skin and fat with constant pressure regardless of the amount of tissue being measured (41, 69, 120). Circumference methods may be added, because they are relatively quick and simple and can yield important chronic disease risk information. For example, waist circumference can assess abdominal fat, and a high waist circumference is associated with an increased risk for type 2 diabetes, high cholesterol, high blood pressure, and certain types of cardiac disease (69).

Anthropometry

Anthropometry, which is the science of measurement applied to the human body, generally includes measurements of height, weight, and selected body girths (69). Ideally, height should be measured with a stadiometer. If a stadiometer is not available, measurement of height requires a flat wall against which the athlete stands, with a measuring tape attached or unattached to the wall. Height is usually measured without shoes to the nearest quarter-inch or half-centimeter (103).

The most accurate body mass or body weight measurement is performed with a certified balance scale, which is generally more reliable than a spring scale and should be calibrated on a regular basis (103). A calibrated electronic scale is an acceptable alternative. Athletes should be weighed while wearing minimal dry clothing (e.g., workout shorts and T-shirt, no shoes). For comparison measurements at a later date, they should dress similarly and be weighed at the same time of day. The most reliable body mass measurements are made in the morning upon rising, after elimination and before ingestion of food or fluids. Level of hydration can result in variability of body mass (weight). Thus, athletes should be encouraged to avoid eating salty food (which increases water retention) the day before weighing and to go to bed normally hydrated.

The most reliable girth measurements are usually obtained with the aid of a flexible measuring tape equipped with a spring-loaded attachment at the end that, when pulled out to a specified mark, exerts a fixed amount of tension on the tape. Girth measurements should be made at the beginning of a training period for comparison with subsequent measurements (69, 103).

Monitoring Protocols, Procedures, and Equipment

When selecting tests for athletic assessment, it is crucial to consider the protocols, procedures, and equipment necessary to effectively implement them. Every athletic environment is unique, requiring careful consideration to optimize these aspects for specific settings.

Protocols are systematic plans outlining guidelines to achieve specific objectives. In athlete testing, they provide a comprehensive framework within which procedures are developed, setting standards for consistency, reliability, and validity (as explained in detail in [chapter 13](#)). A monitoring protocol represents the overarching strategy for conducting performance assessments, detailing the overall approach for tracking an athlete's physical capacities and specifying test types and frequencies throughout the season.

Using Artificial Intelligence to Analyze Testing Data

Artificial intelligence (AI) has emerged as a potentially transformative tool in sport. Yet there is still much confusion, apprehension, and skepticism about the value of AI. AI is defined as "the ability of a digital computer to perform tasks commonly associated with intelligent beings ... such as the ability to reason, discover meaning, generalize, or learn from past experience" (31). Within AI, **machine learning (ML)** is the ability of computer machines to learn from and develop data without human input or interaction. These approaches offer various advantages to sport, including automating or semiautomating data collection, processing data into meaningful insights, identifying crucial information related to health and performance, and creating models that support complex decision making (40). Such models can handle a vast number of factors and their interactions, which can be too complex for humans to process.

In the context of athlete testing, AI holds numerous potential applications. Its capacity to process large volumes of data is particularly suited to biomechanical analysis, leading to its integration into a number of movement analysis solutions (12). Moreover, AI can be used to streamline certain technologies and tests, making jump, speed, and velocity-based gym exercise assessments feasible with just a mobile phone (9). By leveraging the quality of modern phone cameras and applying AI and ML techniques to analyze the data, such tests become accessible to more strength and conditioning professionals. However, it is crucial for practitioners to scrutinize the accuracy and validity of such technologies.

The use of AI in injury prediction is a widely debated subject (12), yet its potential to enhance existing approaches cannot be overlooked. The ability of AI and ML to process vast quantities of data and comprehend complex interactions within and across data sets may further such applications. Nevertheless, practitioners must ensure sufficient data collection quantity and quality. As the maxim "garbage in, garbage out" implies, the success of any model is contingent on the quality of its inputs. AI's application in analyzing athlete testing data is a burgeoning field that holds promise for the future. Given its rapid and constant evolution, it is imperative that practitioners stay informed of developments in this area.

Procedures are detailed, step-by-step instructions on executing specific tasks within the protocol framework. They provide explicit guidance on methods to ensure consistent data collection. In physical performance assessments, procedures detail the steps for conducting the tests outlined in this chapter, such as vertical jumps, aerobic capacity, or agility performance, ensuring standardized methodologies and reliable data.

Selecting appropriate equipment for athlete physical performance tests can be challenging due to the wide range of tools and technologies now available. A decision-making framework for implementing technology in sport (159) suggests four reflective questions:

1. Would the information provided be helpful?
2. Can you trust the information provided?
3. Can you integrate, manage, and analyze the data provided effectively?
4. Can you implement the technology in your practice?

Selecting suitable technology involves ensuring device validity and reliability, considering practicality and ease of use, and confirming relevance to athletes and their specific sport. This comprehensive approach ensures that the assessments are accurate, relevant, and actionable, supporting athlete performance and reduced risk of injury.

In summary, considering protocols, procedures, and equipment prior to commencement of testing is essential. Regularly reviewing these aspects ensures testing batteries remain consistent, relevant, and effective for athletes' training programs.

SELECTED TEST PROTOCOLS AND SCORING DATA

MAXIMUM MUSCULAR STRENGTH (LOW-SPEED STRENGTH)

- 1 1RM Bench Press [304](#)
- 2 1RM Bench Pull [304](#)
- 3 1RM Back Squat [305](#)

MAXIMUM MUSCULAR POWER (HIGH-SPEED STRENGTH)

- 4 Isometric Mid-Thigh Pull [305](#)
- 5 Nordic Hamstring Test [307](#)
- 6 1RM Power Clean [308](#)
- 7 Standing Long (Broad) Jump [308](#)
- 8 Vertical Jump [308](#)
- 9 Squat Jump (Static Vertical Jump) [311](#)
- 10 Drop Jump [312](#)
- 11 Rebound Jump [313](#)
- 12 Margaria-Kalamen Test [314](#)
- 13 Velocity-Based Testing (VBT) [315](#)

ANAEROBIC CAPACITY

- 14 300-Yard (274 m) Shuttle Run [316](#)
- 15 90-Second Box Jump Test [317](#)

LOCAL MUSCULAR ENDURANCE

- 16 Partial Curl-Up [318](#)
- 17 Push-Up [319](#)
- 18 Army Fitness Test (AFT) Hand-Release Push-Up (HRP) [320](#)
- 19 YMCA Bench Press Test [321](#)

AEROBIC CAPACITY

- 20 1.5-Mile (2.4 km) Run [321](#)
- 21 12-Minute Run [321](#)
- 22 Yo-Yo Intermittent Recovery Test [322](#)
- 23 Maximal Aerobic Speed Test [323](#)
- 24 30-15 Intermittent Fitness Test (IFT) [323](#)

CHANGE OF DIRECTION OR AGILITY

- 25 T-Test [324](#)
- 26 Hexagon Test [325](#)
- 27 5-10-5 Test [326](#)
- 28 5-0-5 Test [326](#)

29 Illinois Test [327](#)

SPEED

30 Straight-Line Sprint Tests [327](#)

BALANCE AND STABILITY

31 Balance Error Scoring System (BESS) [328](#)

32 Star Excursion Balance Test (SEBT) [329](#)

FLEXIBILITY

33 Sit-and-Reach Test [330](#)

34 Overhead Squat [331](#)

BODY COMPOSITION

35 Skinfold Measurements [332](#)

36 Bioelectrical Impedance (BIA) [334](#)

ANTHROPOMETRY

37 Girth Measurements [334](#)

Maximum Muscular Strength (Low-Speed Strength)

1 1RM BENCH PRESS

Equipment

- A barbell, weight plates, and two safety locks; enough total weight to accommodate the maximum load of the strongest athlete; and a variety of plate sizes to allow for 5-pound (2.3 kg) gradations in weight
- A sturdy bench press bench with integral bar rack (preferably of adjustable height)

Personnel

- One spotter, one recorder

Procedure

1. Instruct the athlete in proper technique for the flat barbell bench press as described in [chapter 16](#).
2. The spotter stands at the head end of the bench throughout the test to help in raising the bar on a failed attempt and to help the athlete place the bar back on the rack.
3. As with any maximal strength test, the athlete first does a specific warm-up of 5 to 10 repetitions with a light to moderate load.
4. Usually, at least two heavier warm-up sets of 2 to 5 repetitions each are completed before the first actual 1RM attempt.
5. Generally, it is desirable to measure the 1RM within three to five attempts after the warm-up; otherwise, fatigue may detract from the final result.
6. A more detailed step-by-step method for the 1RM protocol is shown in [figure 18.8](#) on [page 546](#).

Note: Normative and descriptive data for the 1RM bench press are presented in [tables 14.1](#) through [14.5](#) near the end of the chapter.

2 1RM BENCH PULL

Equipment

- A barbell, weight plates, and two safety locks; enough total weight to accommodate the maximum load of the strongest athlete; and a variety of plate sizes to allow for 5-pound (2.3 kg) gradations in weight
- A high flat bench (specifically, the *seal row bench* is widely available for this exercise)

Personnel

- One spotter, one recorder

Procedure

1. Instruct the athlete in proper technique for the bench pull ([figure 14.3a](#)).
2. The athlete grasps the bar with a closed pronated grip, wider than shoulder width.
3. Bench height is set so the athlete can use a comfortable grip while the weight is off the ground in the hang position.
4. The athlete starts the lift from the hang position, and the grip should be consistent from test to test.
5. The bar is pulled up toward the lower chest or upper abdomen with the elbows pointed up ([figure 14.3b](#)).
6. The head position can remain either down or to the side but must remain in contact with the bench throughout the test.
7. A valid repetition is one in which the bar touches the underside of the bench and the bar is lowered in a controlled manner to the hang position with full elbow extension without touching the ground.
8. The feet should remain off the ground throughout the test and in the same position throughout.
9. A more detailed step-by-step method for the 1RM protocol is shown in [figure 18.8](#) ([page 546](#)).

Note: Descriptive data for the 1RM bench pull are presented in [table 14.4](#) near the end of the chapter.



FIGURE 14.3 (a) Starting position and (b) top position of the bench pull.

3 1RM BACK SQUAT

Equipment

- A barbell, weight plates, and two safety locks; enough total weight to accommodate the maximum load of the strongest athlete; and a variety of plate sizes to allow for 5-pound (2.3 kg) gradations in weight
- A sturdy squat rack with adjustable spotting bars to support the weight of the bar if the athlete is unable to rise (as an alternative, one spotter can be used at each end of the bar)
- A flat and solid surface to stand on

Personnel

- Two spotters, one recorder

Procedure

1. Instruct the athlete in proper technique for the back squat as described in [chapter 16](#).
2. Warm-up sets are performed as for the 1RM bench press test. However, the loads lifted are typically heavier than in the 1RM bench press test, so the load increments will be greater than those of the 1RM bench press.
3. Refer to [figure 18.8](#) ([page 546](#)) for a 1RM testing protocol.

Note: Normative and descriptive data for the 1RM back squat are presented in [tables 14.1](#) through [14.5](#) near the end of the chapter.

Maximum Muscular Power (High-Speed Strength)

4 ISOMETRIC MID-THIGH PULL

The isometric mid-thigh pull (IMTP) is a multijoint maximal isometric strength test.

Equipment

- Force plate (or two adjacent force plates) that samples at a minimum of 1,000 Hz, but it can also be conducted using a strain gauge or crane scale that samples at 100 to 133 Hz (29, 32)
- An immovable fixed bar to measure peak force and time-dependent variables such as the rate of force development; this specific testing rack is designed for isometric testing and allows for stepwise adjustments to the bar height. If a strain gauge is used, a handle is attached to the strain gauge with a chain.
- Weightlifting straps
- Computer or tablet with specialized software that allows for the collection and analysis of force–time curve data
- Goniometer to record the knee and hip angles

Personnel

- One tester/recorder



FIGURE 14.4 Isometric mid-thigh pull (IMTP) test position.

Procedure

1. The testing rack needs to be set to a height that allows the athlete to mirror the “second pull” position used in weightlifting (i.e., clean or power clean). In this position the immovable bar is positioned across the upper portion of the thigh, immediately inferior to the pelvis (58) ([figure 14.4](#)).
2. When performing the test, the athlete should have an upright trunk position (62, 63) with the feet roughly hip-width apart and a slight flex of the knee while grasping the bar with a clean grip while using weightlifting straps (58). In this position the knee angle should be approximately 120 degrees and 145 degrees, and the hip angle should be approximately 125 degrees and 150 degrees (62). However, it is important to note that the exact bar position as well as knee and hip angles are difficult to recommend because an individual’s anthropometrics will exert a large influence on the exact position during the test.
3. Before conducting the isometric mid-thigh pull test it is important that between one and three familiarization sessions are performed in order to enhance the reliability of the test (62). Less familiarization is needed with trained athletes, especially those who train regularly with weightlifting derivatives. First, the athlete should complete a dynamic warm-up that includes dynamic stretches (e.g., bodyweight squats and lunges) as well as dynamic lifting tasks (e.g., low-load and moderate-load mid-thigh pulls). After this is completed, a series of progressively intense isometric actions should be performed at perceived intensities of 50%, 70%, and 90% of maximum, with 1 to 2 minutes rest between trials (29, 62). During this time the strength

and conditioning professional should monitor the athlete's body posture and make corrections as needed.

4. After completing the last submaximal trial, the athlete should complete a minimum of three maximum-effort trials separated by 1 to 2 minutes of rest. If the main goal of the test is to determine the peak force that the athlete can produce, each trial should last between 3 and 5 seconds. If the rate of force development (RFD) is of interest, each trial should last up to 1 second (54). Regardless of which protocol is used, if the peak force between trials varies by more than 15%, additional trials are warranted (105).
5. Regardless of the protocol used to collect the force–time curve, all trials need to be visually screened to determine whether they meet preestablished criteria (29). If the strength and conditioning professional notes any of the following, the trial should be discarded and another trial should be conducted:
 - A quiet standing period of 1 s with >50 N change in force
 - Pre-tension (<50 N) while maintaining a rigid torso before initiating force application
 - A countermovement prior to the onset of force application
 - Peak force at the end of the pull
6. If the trial is deemed acceptable but the peak force of each trial is not within 15% of the others, additional trials should be conducted.
7. When analyzing the results of an IMTP test, the most common measures include the peak force and the force at specific time points (i.e., 50, 100, 150, 200, and 250 ms) ([figure 14.5](#)). These metrics are often reported in absolute and relative terms (i.e., force per kg body weight). Time-dependent variables such as the RFD and impulse are often quantified (and reflect the change in force production during a certain time period as represented by Δ). The peak RFD is typically quantified with a 20-millisecond moving window, while the RFD and impulse are quantified during predetermined time windows (i.e., 0-50, 0-100, 0-200, and 0-250 ms).

Note: Normative data for the IMTP are presented in [table 14.6](#) near the end of the chapter.

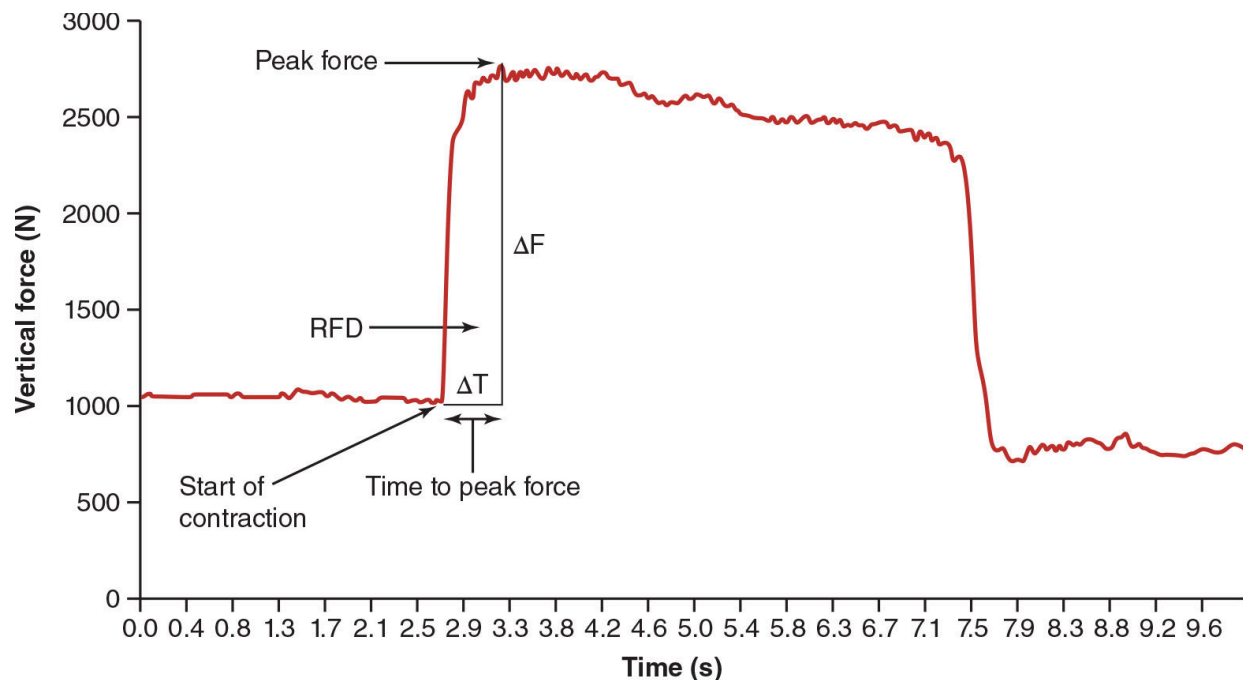


FIGURE 14.5 IMTP force-time curve.

Reprinted by permission from C.J. Brady, A.J. Harrison, and T.M. Comyns, "A Review of the Reliability of Biomechanical Variables Produced During the Isometric Mid-Thigh Pull and Isometric Squat and the Reporting of Normative Data," *Sports Biomechanics* 19, no. 1 (2020): 1-25, reprinted by permission of the publisher (Taylor & Francis Ltd, <http://www.tandfonline.com>).

5 NORDIC HAMSTRING TEST

Equipment

- A commercially available instrumented device for the measurement of eccentric hamstring strength or a video recording device
- A stable flooring surface on which the dynamometer should be placed

Personnel

- One tester/recorder

Procedure

1. Set up the instrumented device on a stable surface. The athlete should kneel on the testing device and place the ankles under the hooks (within which force cells are housed to quantify force). The athlete should be careful to only contact the hooks with the back of the ankles and to not push off the board with the toes in order to gain leverage. If not using an instrumented device but using videography instead, the camera (or mobile phone) should be placed in the sagittal view and fixed using a tripod or similar and positioned so that the full range of motion is recorded. The knees of the athlete should be a 90-degree angle of flexion relative to the lower leg and the rest of the body ([figure 14.6a](#)). Care should be taken to ensure that the athlete does not flex forward at the hips, because occasionally athletes will do this in order to reduce the eccentric loading on the hamstrings.
2. The athlete should then lean forward under control (i.e., controlled acceleration) by extending the knees, increasing the knee angle, and maintaining a straight line from the knees through the shoulders ([figure 14.6b](#)).
3. After lowering to the ground in this fashion, the athlete may then use the hands to push back up to the starting position. Typically, 3 to 6 sets of this test are performed.

4. If using an instrumented device, record absolute force of eccentric hamstring force, typically measured in newtons (N). Ratio or allometric scaling of peak force, as discussed earlier in the chapter, may also be considered with Nordic hamstring force outputs to account for variations in body mass and length on the peak force measurements. Other metrics such as torque in newton-meters (Nm) or the percentage of asymmetry present between left and right hamstrings may also be recorded if desired. If using videography alone, the break point angle (the angle at which the external knee flexor moment can no longer be resisted and the athlete loses active control of downward motion) can be reported as a moderate proxy of eccentric hamstring strength.

Note: Descriptive data for the Nordic hamstring test are presented in [table 14.7](#) near the end of the chapter.

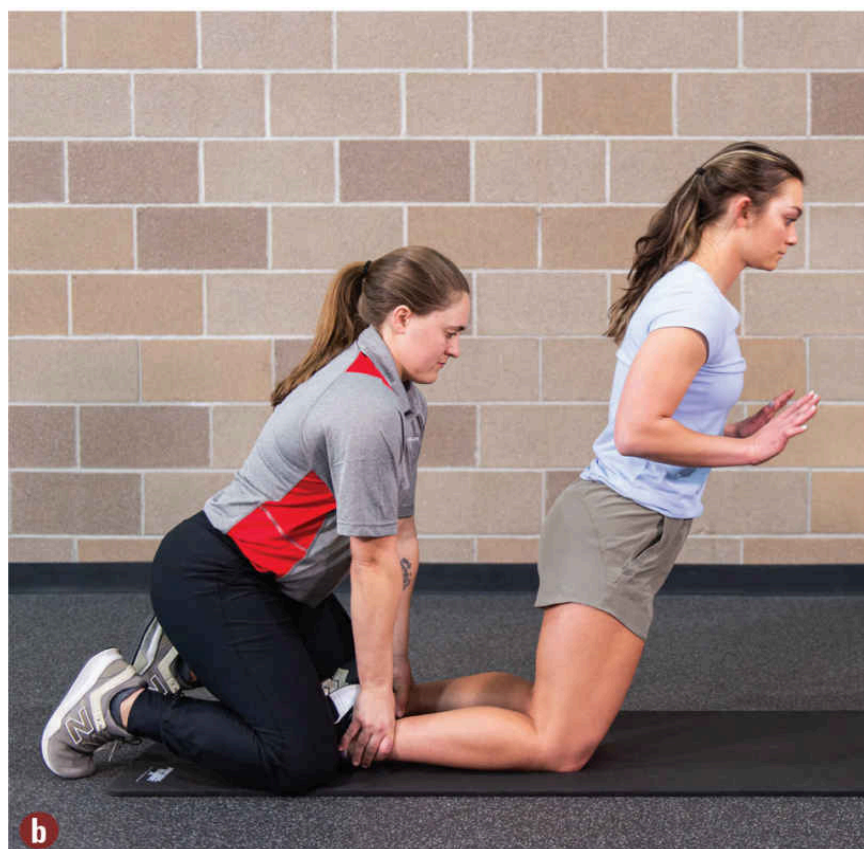
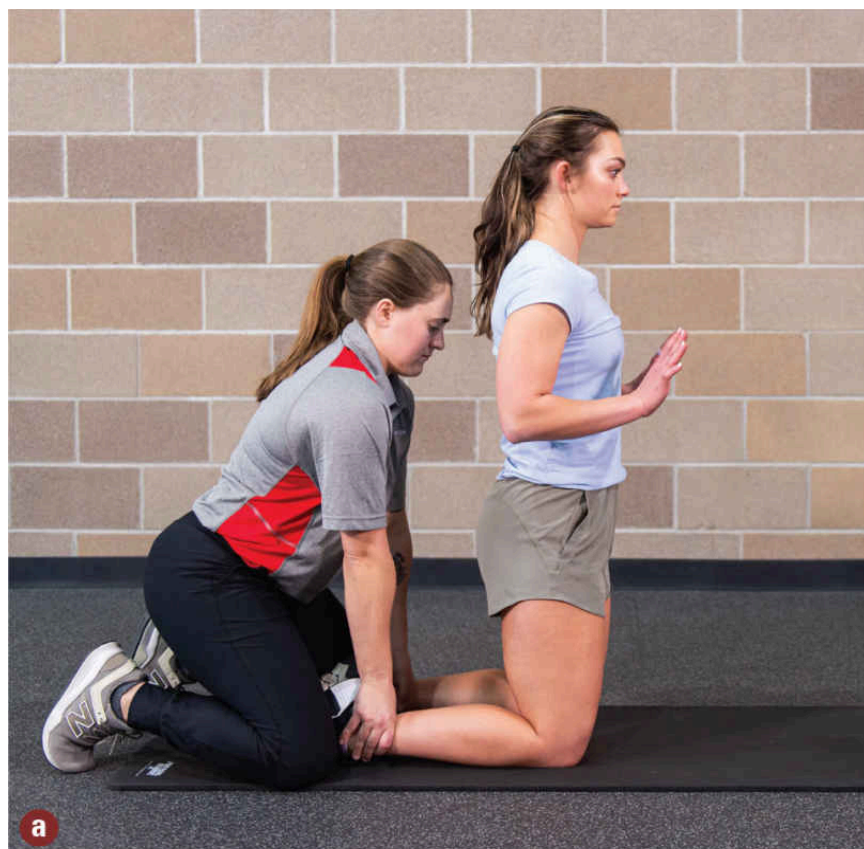


FIGURE 14.6 Nordic hamstring test (without an instrumented device).

6 1RM POWER CLEAN

Note: Because the power clean exercise has high technical demands, two athletes with the same muscular power capacity can differ greatly in their tested 1RM, lessening the value of the test for predicting athletic performance.

Equipment

- An Olympic-style barbell with a revolving sleeve, weight plates, and two safety locks; enough total weight to accommodate the maximum load of the strongest athlete; and a variety of plate sizes to allow for 5-pound (2.3 kg) gradations in weight
- A lifting platform or designated area set apart from the rest of the facility for safety

Personnel

- One tester/recorder

Procedure

1. Instruct the athlete in proper technique for the power clean as described in [chapter 16](#).
2. Warm-up sets are performed and load increments are selected as for the 1RM bench press test.
3. Refer to [figure 18.8 \(page 546\)](#) for a 1RM testing protocol.

Note: Normative and descriptive data for the 1RM power clean are presented in [tables 14.1 through 14.4](#) near the end of the chapter.

7 STANDING LONG (BROAD) JUMP

Equipment

- A flat jumping area at least 20 feet (6 m) in length, which can be a gym floor, artificial turf, grass field, or a track
- A tape measure at least 10 feet (3 m) long
- Duct tape or masking tape
- Permissible alternative: a commercial jumping mat premarked in half-inch (1 cm) increments

Personnel

- One distance judge, one recorder

Procedure

1. Place a 2- to 3-foot (0.6-0.9 m) length of tape on the floor to serve as a starting line.
2. The athlete stands with the toes just behind the starting line.
3. The athlete performs a countermovement and jumps forward as far as possible.
4. The athlete must land on the feet for the jump to be scored. Otherwise, the trial is repeated.
5. A marker is placed at the back edge of the athlete's rearmost heel, and the tape measure determines the distance between the starting line and the mark.
6. The best of three trials is recorded to the nearest 0.5 inch or 1 centimeter.

Note: Normative and descriptive data for the standing long jump are presented in [tables 14.8 to 14.10](#) near the end of the chapter.

8 VERTICAL JUMP

Equipment

- A force plate set up on a stable surface
- A suitable computer, tablet, or cell phone connected to force plate hardware and with the corresponding software installed
- Permissible alternatives: a smooth wall with a ceiling higher than the highest jumper's jump height plus chalk of a different color than the wall or a commercial device for vertical jump testing (e.g., Vertec)

Personnel

- One tester/recorder

Procedure (Using a Force Plate)

1. Before testing, the tester should calibrate and set up the force plate as per the manufacturer's instructions.
2. The force plate should always be set up on a stable surface (ideally in the same location each time) and the force zeroed with nothing on the plates immediately before the athlete tests. Without a preparatory or stutter step, the athlete performs a countermovement by quickly flexing the knees and hips, moving the trunk forward and downward, and swinging the arms backward.
3. The countermovement jump is most commonly conducted with hands in a fixed position on the hips throughout the takeoff portion of the jump to isolate the power and coordination capacity of the lower limbs. Jumping with the use of an arm swing can also be tested and is referred to as an *Abalakov jump*. Either way, consistency should be used and each jump should be tracked as to whether it was performed with (Abalakov) or without an arm swing.
4. [Figure 14.7](#) shows an example of how the vertical ground reaction forces change across the five phases of a vertical jump with a countermovement when measured with a force plate testing system.

Procedure (Using a Wall and Chalk)

1. The tester rubs chalk on the fingertips of the athlete's dominant hand.
2. The athlete stands with the dominant shoulder about 6 inches (15 cm) from the wall and, with both feet flat on the floor, reaches as high as possible with the dominant hand and makes a chalk mark on the wall.
3. The athlete then lowers the dominant hand and, without a preparatory or stutter step, performs a countermovement by quickly flexing the knees and hips, moving the trunk forward and downward, and swinging the arms backward. During the jump, the dominant arm reaches upward while the nondominant arm moves downward relative to the body.
4. At the highest point in the jump, the athlete places a second chalk mark on the wall with the fingers of the dominant hand, using a swiping motion of the fingers. The score is the vertical distance between the two chalk marks.
5. The best of three trials is recorded to the nearest 0.5 inch or 1.0 centimeter.

Procedure (Using a Commercial Vertec Device)

1. The tester adjusts the height of the stack of movable color-coded horizontal plastic vanes to be within the athlete's standing reach height. The highest vane that can be reached and pushed forward with the dominant hand while the athlete stands flat-footed determines the standing touch height.
2. The vane stack is then raised by a measured distance (marked on the shaft holding the vanes) so that the athlete will not jump higher or lower than the set of vanes. This requires a rough

estimate of how high the particular athlete will jump, but a correction can be made on the second attempt if necessary.

3. Without a preparatory or stutter step, the athlete performs a countermovement by quickly flexing the knees and hips, moving the trunk forward and downward, and swinging the arms backward ([figure 14.8a](#)). During the jump, the dominant arm reaches upward while the nondominant arm moves downward relative to the body.
4. At the highest point in the jump, the athlete taps the highest possible vane with the fingers of the dominant hand ([figure 14.8b](#)). The score is the vertical distance between the height of the highest vane tapped during the standing vertical reach and the vane tapped at the highest point of the jump.
5. The best of three trials is recorded to the nearest 0.5 inches or 1 centimeter (the distance between adjacent vanes).

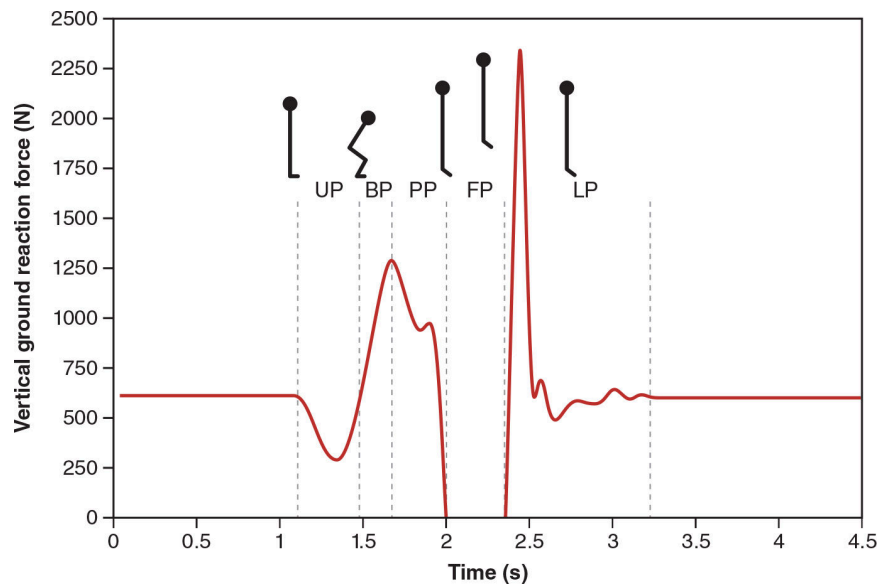


FIGURE 14.7 Example force–time curve of a countermovement jump when measured with a force plate testing system. Vertical dotted lines divide the typical phases (UP, unloading phase; BP, breaking phase; PP, propulsive phase; FP, flight phase; LP, landing phase).

Reprinted by permission from Z. Anicic, D. Janicijevic, O.M. Knezevic, et al., "Assessment of Countermovement Jump: What Should We Report?," *Life* 13, no. 1 (2023): 190. <https://doi.org/10.3390/life13010190>. Distributed under the terms of the Creative Commons Attribution 4.0 International License (<http://creativecommons.org/licenses/by/4.0/>).

The strength and conditioning professional should be aware of potential sources of error with the jump-and-reach-style tests. These include a disconnect between the moment when the athlete reaches the wall or vanes with the hand and the moment the maximum height of the body's center of mass is achieved; additionally, shoulder mobility issues may limit some athlete's reach height. Also, jump height values calculated using common technologies such as force plates and contact mats are most often measured as the maximum total body center of mass height relative to its height at takeoff and not relative to its height when standing still (158). Thus, jump height values from jump-and-reach tests should not be compared directly to, or used interchangeably with, jump height values recorded with such technologies or, indeed, with different jump techniques, such as when keeping arms akimbo.

Note: Descriptive data for the vertical jump are presented in [table 14.5](#) and [table 14.10](#) near the end of the chapter.

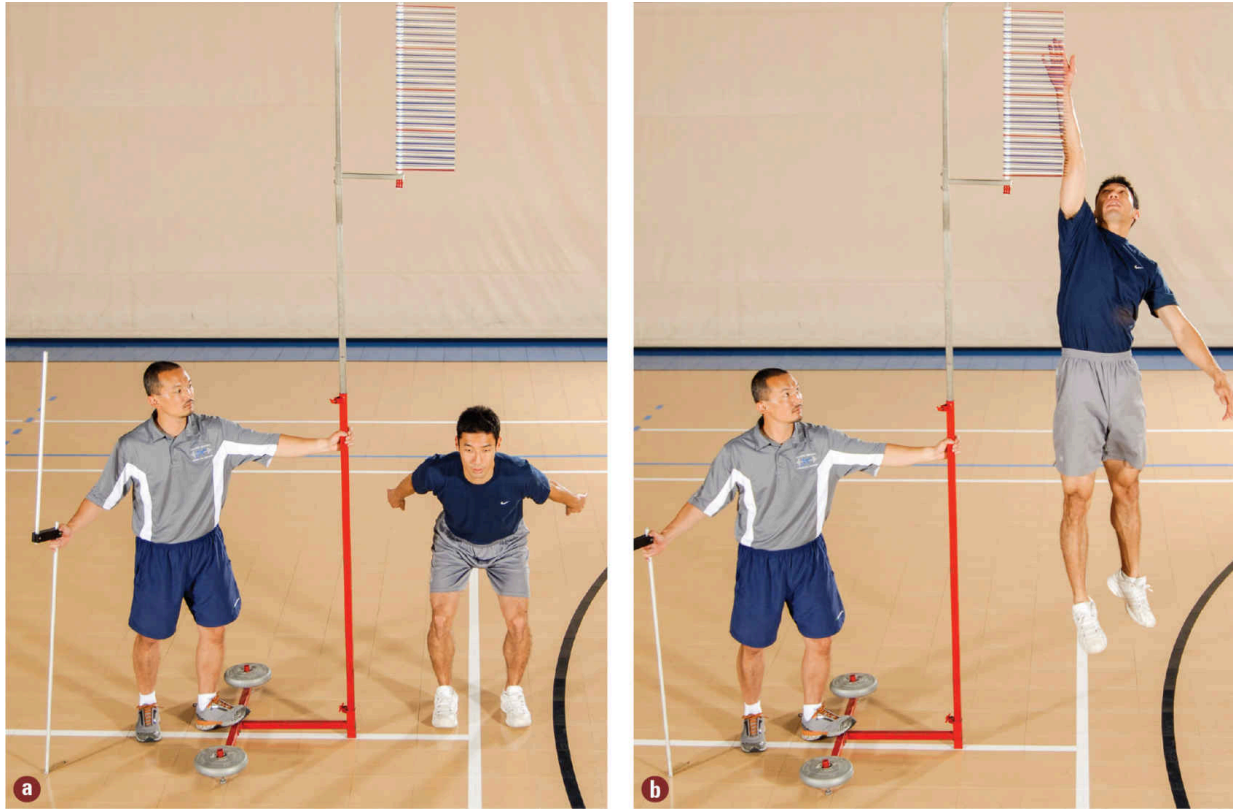


FIGURE 14.8 (a) Starting position and (b) maximum height of the vertical jump using the Vertec.

9 SQUAT JUMP (STATIC VERTICAL JUMP)

The equipment needed and the personnel required for the squat (static) vertical jump are the same as the vertical jump with a countermovement.

Procedure

1. The test procedures are essentially the same as for the vertical jump, except that the countermovement is removed. Begin with the athlete standing on a calibrated, zeroed force platform. (*Note:* The squat jump and vertical jump with countermovement can also be tested using a switch or contact mat system ([figure 14.9](#)) [158], which estimates jump height [51] but cannot measure force outputs.)
2. The athlete descends into a squat position (knee angle approximately 110 degrees) and holds this position for 2 to 3 seconds before jumping vertically, keeping hands fixed on hips.
3. The takeoff and landing positions, as well as jumping strategy, should be the same for each trial. Specifically, the tester should ensure there is no countermovement (dip) at the start of the jump, both through visual observation and, where a force plate is used, by assessing the force–time curve.
4. The best of three trials is recorded. The ratio of the vertical jump height with countermovement to squat jump height can be calculated as the eccentric utilization ratio (99).
5. [Figure 14.10](#) shows an example of how the vertical ground reaction forces change across the three phases of a static vertical jump when measured with a force plate testing system.

Note: Descriptive data for the static vertical jump are presented in [table 14.10](#) near the end of the chapter.

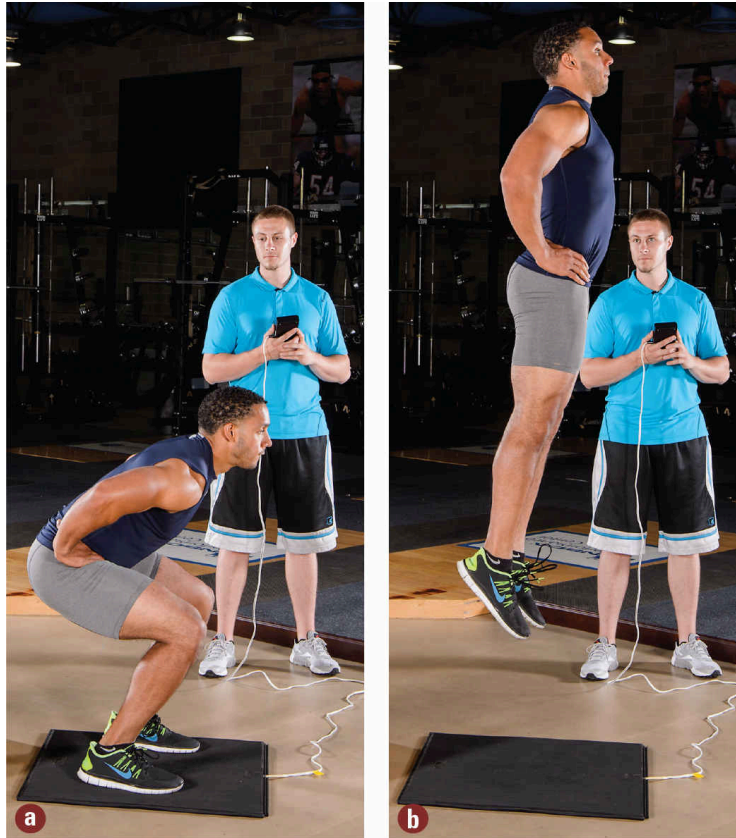


FIGURE 14.9 (a) Starting position and (b) maximum height of the static vertical jump when measured with a switch mat.

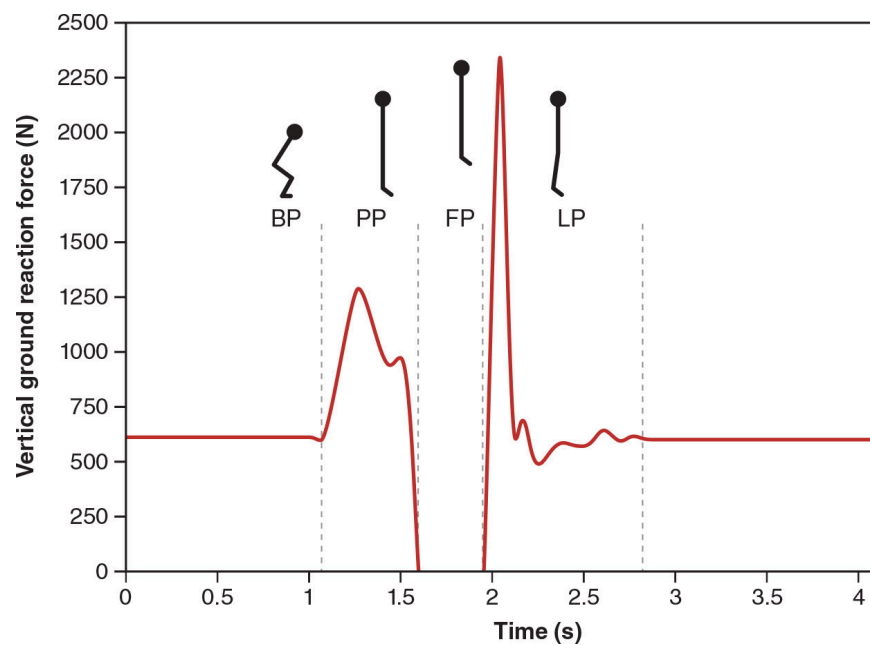


FIGURE 14.10 Example force–time curve of a static vertical (squat) jump when measured with a force plate testing system. Vertical dotted lines divide the typical phases (BP, breaking phase; PP, propulsive phase; FP, flight phase; LP, landing phase).

Adapted from Z. Anicic, D. Janicijevic, O.M. Knezevic, et al., "Assessment of Countermovement Jump: What Should We Report?," *Life* 13, no. 1 (2023): 190. <https://doi.org/10.3390/life13010190>. Distributed under the terms of the Creative Commons Attribution 4.0 International License (<http://creativecommons.org/licenses/by/4.0/>).

10 DROP JUMP

Equipment

- A box that may vary in height from 8 to 16 inches (20-40 cm)
- Floor with good traction
- Measurement equipment such as force plates (with corresponding software on laptop or tablet), switch mats, optoelectronic systems, and smartphone applications

Personnel

- One tester/recorder

Procedure

1. When performing a drop vertical jump, the athlete should step out before free-falling from as close as possible to the height of the box rather than stepping down (i.e., lowering) or jumping off (i.e., rising) the box. Box height selection for an athlete can vary but is typically 20 centimeters (8 in.), 30 centimeters (12 in.), or 40 centimeters (16 in.) ([figure 14.11a](#)). The higher the box, the greater the plyometric demand. When using a device that is not flush with the ground or base of the box when conducting the drop vertical jump test, such as when using a switch mat, the strength and conditioning professional should measure the box height from the top of the box to the top surface of the switch mat.
2. The athlete steps out from the box and lands on the floor, force plate, or switch mat ([figure 14.11b](#)) and, as quickly as possible, jumps again ([figure 14.11c](#)). Drop jumps should be performed with the athlete attempting to jump with a short ground contact time while still aiming to jump as high as possible. A short ground contact time is achieved by limiting flexion of the ankles, knees, and hips while in contact with the ground by adopting high leg stiffness (high force with shallow range of motion). It is advised that at least 2 to 3 sets are performed.
3. The most reported performance variables following drop vertical jump testing are jump height (from flight time), ground contact time, and **reactive strength index** (calculated as jump height divided by ground contact time) (91). The reactive strength index is often considered an indication of how well an athlete can recruit the fast component of the stretch-shortening cycle.
4. [Figure 14.12](#) shows an example of the force–time curve with labels pointing out the peak vertical ground reaction force, contact time, and flight time when measured with a force plate testing system.

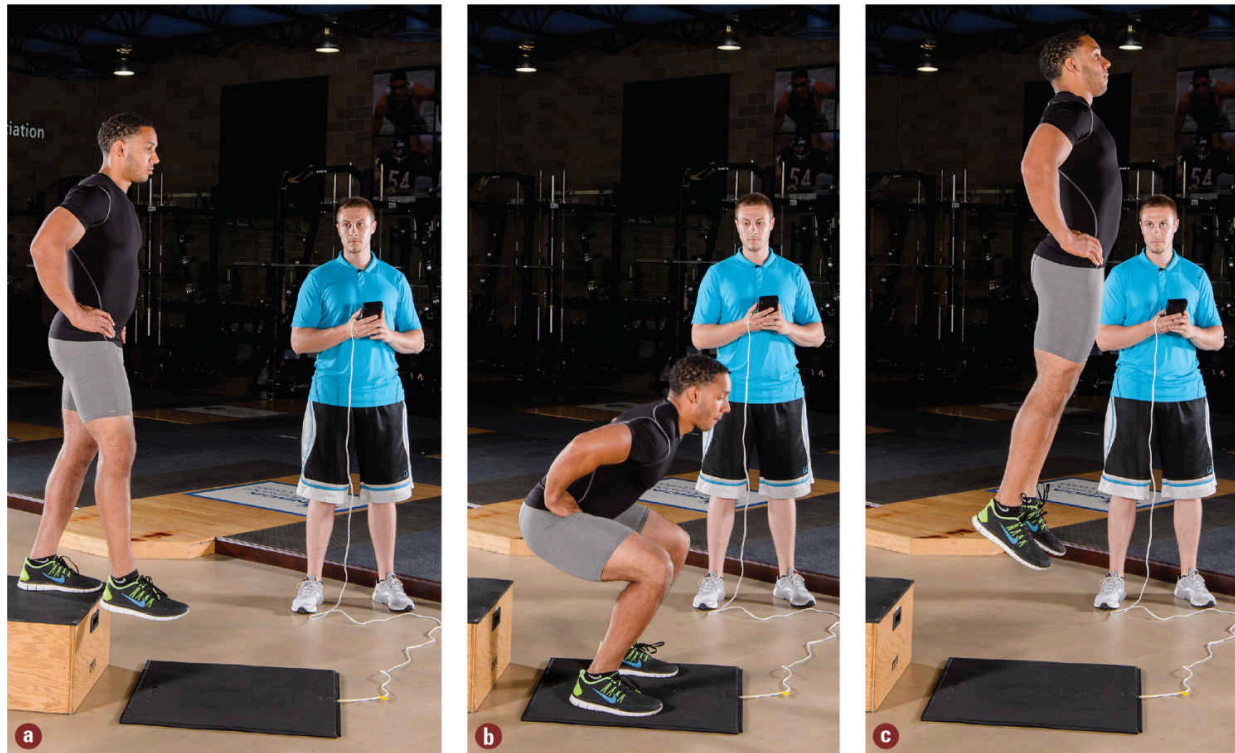


FIGURE 14.11 (a) Starting position on box, (b) contact on switch mat, (c) maximum height of the drop jump to measure reactive strength index.

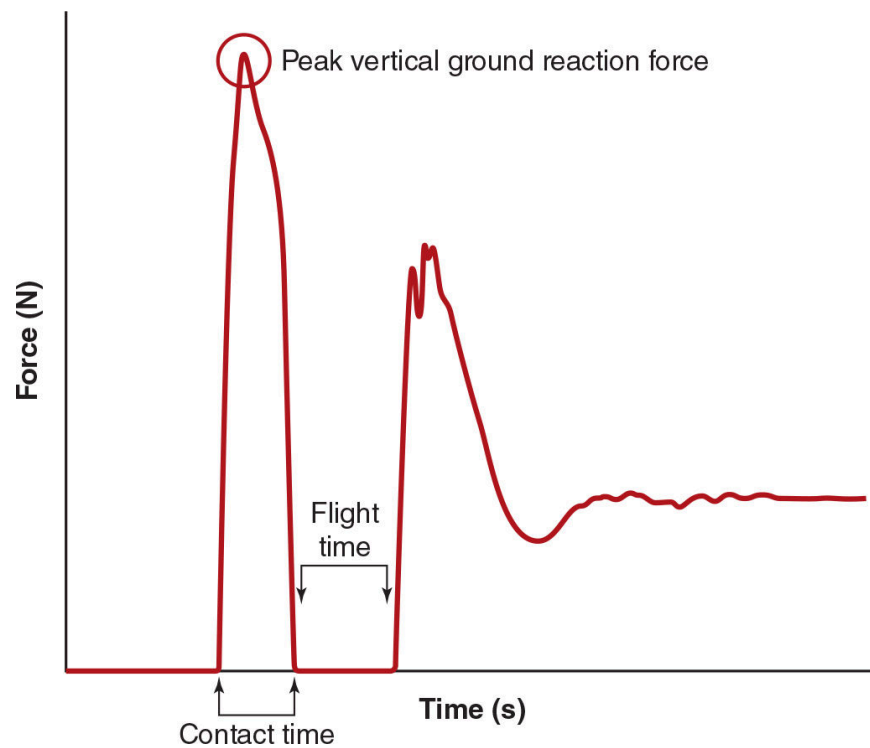


FIGURE 14.12 Example force-time curve of a drop jump.

Reprinted by permission from R. Healy and A.J. Harrison, "The Effects of a Unilateral Gluteal Activation Protocol on Single Leg Drop Jump Performance," *Sports Biomechanics* 13, no. 1 (2014): 33-46, reprinted by permission of the publisher (Taylor & Francis Ltd, <http://www.tandfonline.com>).

11 REBOUND JUMP

Equipment

- Floor with good traction
- Measurement equipment such as force plates (with corresponding software on a laptop or tablet), jump mats, optoelectronic systems, and smartphone applications

Personnel

- One tester/recorder

Procedure

1. Rebound-based vertical jumps require the athlete to jump from a preceding vertical jump. When performing a rebound vertical jump, the athlete should first complete a maximal-effort vertical countermovement jump followed by either a predetermined number of rebound vertical jump repetitions or a predetermined set duration (e.g., 10 seconds). Rebound vertical jumps from a ground-start position have been performed with single or multiple repetitions, including the 10-5 repeated jump test (64), which requires 10 maximal vertical rebound jumps following the initial maximum countermovement jump.
2. All styles of rebound vertical jump should be performed with the athlete attempting to jump with a short ground contact time while still aiming to jump as high as possible. A short ground contact time is achieved by limiting flexion of the ankles, knees, and hips while in contact with the ground by adopting high leg stiffness (high force with shallow range of motion).
3. As per the drop jump, the most reported performance variables following rebound vertical jump testing are jump height, ground contact time, and reactive strength index (calculated as jump height divided by ground contact time). For the 10-5 repeated jump test, the best 5 repetitions in terms of reactive strength index are used and averaged for the test outcome.

Note: Descriptive data for the rebound jump are presented in [table 14.11](#) near the end of the chapter.

12 MARGARIA-KALAMEN TEST

Equipment

- Staircase with nine or more steps, each approximately 7 inches (18 cm) high, and a straight and flat lead-up area 20 feet (6 m) or more in length ([figure 14.13](#))
- Measuring tape or stick
- An electronic timing system with both a start and a stop switch mechanism
- Scale

Personnel

- One tester/recorder

Procedure

1. The height of each step is measured with a ruler or tape measure, and the elevation from the third step to the ninth step is calculated ($6 \times \text{step height}$).
2. The timer start switch mechanism is placed on the third step, and the stop switch mechanism is placed on the ninth step.
3. The athlete to be tested is weighed on a scale, warms up, and practices running up the stairs three steps at a time.
4. When ready, the athlete sprints toward the stairs from a standing start 20 feet (6 m) from the base of the stairs and then up the staircase three steps at a time (third step to sixth step to

ninth step) as fast as possible.

5. The time from third- to ninth-step contact is determined to the nearest 0.01 second using the timing system.
6. Power in watts is calculated as the athlete's weight (w) in newtons ($\text{lb} \times 4.45$ or $\text{kg} \times 9.807$) times height (h) in meters ($\text{inches} \times 0.0254$) from the third step to the ninth step divided by the measured time interval (t) in seconds; $P(\text{watts}) = (w \times h) / t$.
7. Repeat the test two more times with a 2- to 3-minute recovery period between each trial.

Note: Normative data for the Margaria-Kalamen test are presented in [table 14.12](#) near the end of the chapter.

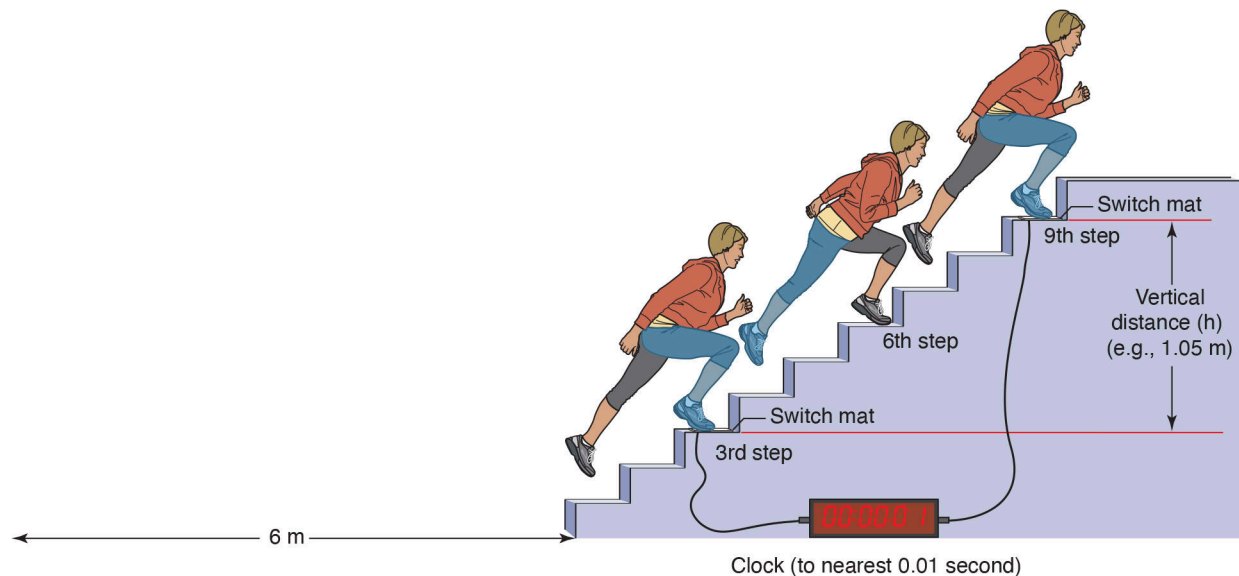


FIGURE 14.13 Margaria-Kalamen stair sprint test.

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13 VELOCITY-BASED TESTING (VBT)

Equipment

- A commercial VBT device and the equipment required to perform the exercise in question (e.g., an inertial measurement unit [IMU], a linear position transducer, a depth-sensing camera system, and a computer-vision enabled smartphone application)
- An Olympic-style barbell with weight plates and locking collars; the weight should be appropriate for the individual athlete being tested (in this case, a 1RM back squat will be the exercise evaluated)

Personnel

- One tester/recorder

Procedure

1. Place the VBT device in a suitable position based on the type of data collection. For example, a linear position transducer will be placed under the barbell and tethered to the barbell, whereas an IMU sensor will be affixed directly to the barbell.
2. The exercise selected (e.g., bench press, squat, deadlift) will depend on the needs of the athlete, individual sport, and training session being performed. In the case of a back squat,

follow the technique described in [chapter 16](#).

3. From the measuring device, obtain the velocity of the barbell typically in meters per second, power measured in watts, and peak forces measured in newtons. Metrics may vary depending on the device used for testing or the individual purposes of the tester and athlete. As outlined earlier in the chapter, VBT data can be used to create a load–velocity profile.

Note: Descriptive data for VBT are presented in [table 14.13](#) near the end of the chapter.

Anaerobic Capacity

14 300-YARD (274 M) SHUTTLE RUN

Equipment

- A stopwatch with at least 0.1-second resolution
- Two parallel lines 25 yards (23 m) apart on a flat surface ([figure 14.14](#))

Personnel

- One timer, two line judges

Procedure

1. Pair off athletes of similar ability.
2. Position two athletes immediately behind one line, facing the other line.
3. On an auditory signal, the athletes sprint to the line 25 yards (23 m) away, making foot contact with it, then immediately sprint back to the first line. Six such round trips are made as fast as possible without stopping ($6 \times 50 \text{ yd} = 300 \text{ yd}$ [274 m]).
4. On completion of the first trial, record both athletes' times to the nearest 0.1 second and start a clock to time a 5-minute rest interval. As each pair of athletes completes the first trial, they may walk and stretch but must stay alert for the starting time on the second trial.
5. After the rest period, the pair of athletes does another trial.
6. The average of two trials is recorded to the nearest 0.1 second.

Note: Descriptive data for the 300-yard (274 m) shuttle are presented in [table 14.14](#) near the end of the chapter.

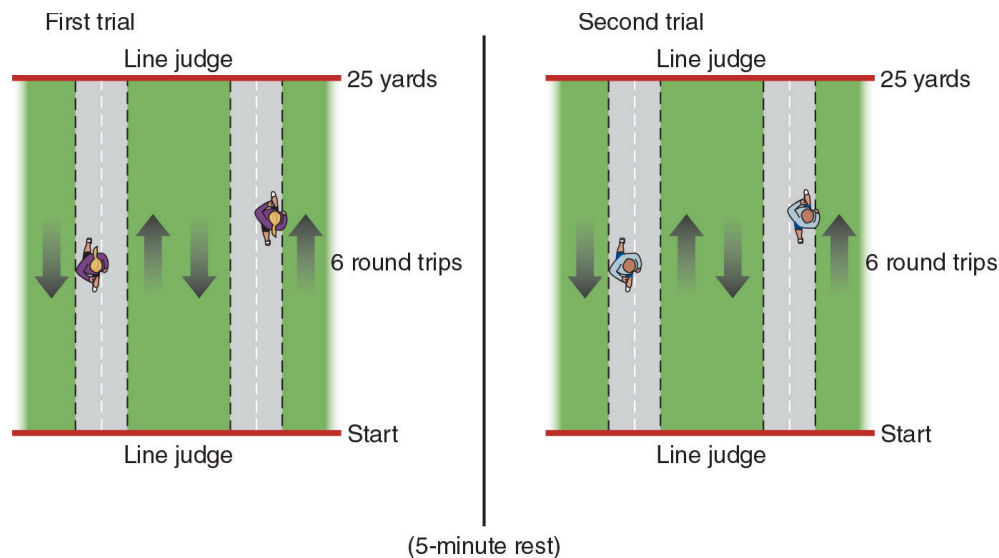


FIGURE 14.14 Ground layout for the 300-yard (274 m) shuttle.

Adapted by permission from G.M. Gillam and M. Marks, "300 Yard Shuttle Run," *National Strength and Conditioning Association Journal* 5, no. 5 (1983): 46.

15 90-SECOND BOX JUMP TEST

Equipment

- Stopwatch
- A box with the dimensions of 18 inches (46 cm) in height and 30 inches (76 cm) in width
- A nonslip surface on which to perform the test in which the athlete can safely land and jump

Personnel

- One recorder/tester

Procedure

1. The purpose of this test is to observe how many times the athlete can jump over the box in a left to right fashion over a duration of 90 seconds. The athlete begins on top of the box and then jumps down laterally to one side before jumping back up to the top of the box again. The athlete then repeats this procedure in the opposite direction. Each time the athlete reaches the top of the box counts as 1 repetition.
2. The athlete must leave the ground with two feet and touch the top of the box with two feet before jumping down to the other side. The athlete is not permitted to step up onto the box.
3. This is repeated for as many times as the athlete is able to during the 90-second duration of the test. The tester should record the number of hops taken with proper form over the course of the test.

Note: Descriptive data for the 90-second box jump test are presented in [table 14.15](#) near the end of the chapter.

Local Muscular Endurance

16 PARTIAL CURL-UP

The partial curl-up test measures the muscular endurance of the abdominal muscles. It is favored over the sit-up test because it eliminates the use of the hip flexor muscles.

Equipment

- Metronome
- Ruler
- Masking tape
- Mat

Personnel

- One recorder/technique judge

Procedure

1. The athlete assumes a supine position on a mat with a 90-degree knee angle ([figure 14.15a](#)). The arms are at the sides, resting on the floor, with the fingers touching a 4-inch-long (10 cm) piece of masking tape positioned perpendicular to the fingers. A second piece of masking tape is situated parallel to the first tape at a distance determined by the age of the athlete (4.7 inches [12 cm] for those younger than 45 and 3.1 inches [8 cm] for those 45 or older).
2. Set a metronome to 40 beats/min and have the athlete do slow, controlled curl-ups to lift the shoulder blades off the mat (trunk makes a 30-degree angle with the mat; [figure 14.15b](#)) in time with the metronome (20 curl-ups per minute). The upper back must touch the floor before each curl-up. The athlete should avoid flexing the neck to bring the chin close to the chest.
3. The athlete performs as many curl-ups as possible without pausing, to a maximum of 75.

Note: Normative data for the partial curl-up are presented in [table 14.16](#) near the end of the chapter.

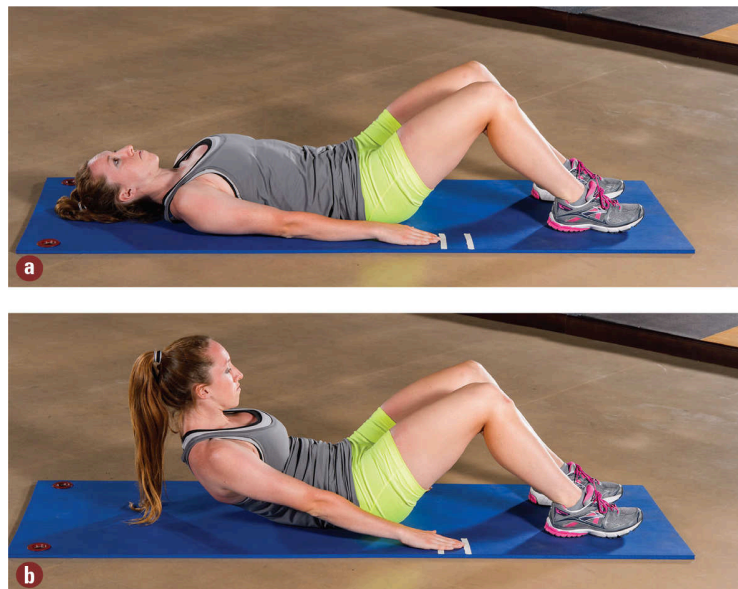


FIGURE 14.15 (a) Beginning position and (b) end position of the partial curl-up test.

17 PUSH-UP

Equipment

- A 4-inch (10 cm)-diameter foam roller (for the modified push-up)

Personnel

- One recorder/technique judge

Procedures

1. Assume the standard push-up starting position with the hands shoulder-width apart and elbows and body straight ([figure 14.16a](#)). The bottom position for the standard push-up is when the chest makes contact with the recorder's fist held vertically against the ground (not shown in [figure 14.16b](#)).
2. A modified version establishes a starting position with the knees rather than the feet contacting the ground; the knees should be flexed to 90 degrees with the ankles crossed ([figure 14.17a](#)). There is no standard criterion for the bottom position of the modified push-up (44), but it has been suggested that the torso should come in contact with a foam roller rather than a fist ([figure 14.17b](#)).
3. For either version, repetitions that do not achieve the bottom position are not counted.

Note: Normative data for the push-up are presented in [table 14.17](#) near the end of the chapter.



FIGURE 14.16 (a) Beginning position and (b) down position of the standard push-up.

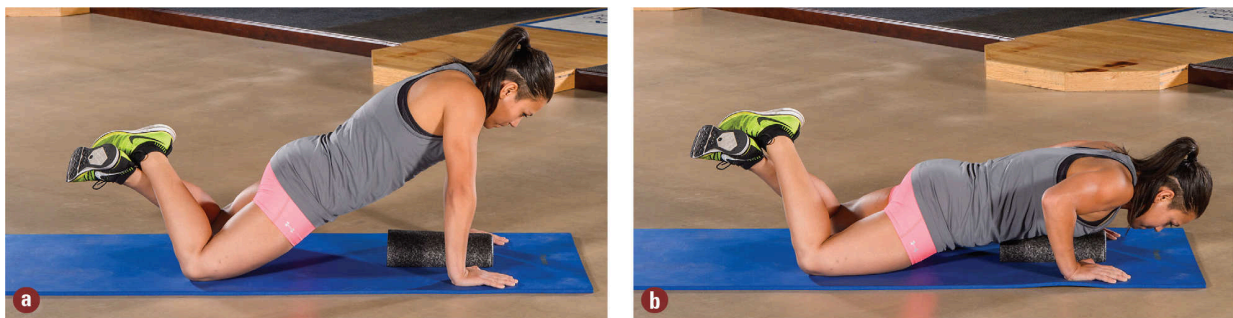


FIGURE 14.17 (a) Beginning position and (b) down position of the modified push-up.

18 ARMY FITNESS TEST (AFT) HAND-RELEASE PUSH-UP (HRP)

Equipment

- Level nonslip ground or flooring surface

Personnel

- One recorder/technique judge

Procedure

1. The starting position of this test will begin with the athlete lying prone on the ground with the chest, hips, and legs touching the ground ([figure 14.18a](#)). Hands are flat on the ground (rather than what is shown in the photo) with the index fingers inside of the shoulders. The feet should be no more than one boot-width apart, and the toes should be touching the ground.
2. The tester will begin the test on the command of "Go," and the athlete must perform as many push-ups as physically able in a 2-minute time period. The athlete should keep a straight line from the ankles to the head ([figure 14.18b](#)). It is not permissible to flex the knees or hips or to lift the feet off of the ground at any point during the test. After the athlete has fully extended the arms, he or she should lower the body back to the ground, allowing the chest, hips, and legs to contact the ground simultaneously. The tester should not count any repetitions in which a straight alignment of the body from head to feet is not maintained.
3. At the point at which the athlete's body comes back into contact with the ground, two variants of this test are acceptable: the hand lift and the arm extension. If the hand lift variant is to be performed, the athlete should lift both hands so as to create visible space between the palms and the ground ([figure 14.18c](#)). The athlete should then return the hands to the starting position and begin another repetition. With the arm extension variant, once the athlete's body has returned to the ground, the athlete will then extend the arms fully to form a 90-degree angle between the arms and the torso ([figure 14.18d](#)). The athlete should then return to the starting position and begin another repetition.
4. The athlete is not allowed to rest at any time during the test with the exception of the front rest position, which consists of the top position of the push-up with the arms fully extended and the body straight. If the tester observes that the athlete has remained motionless for more than 5 seconds, the test should be discontinued and the repetitions performed to this point counted as the athlete's score for the test. The test will also be discontinued if the athlete lifts a foot off of the ground or touches a knee to the ground. Any breakdown in technique such as flexing the hips or knees should not be counted in the repetition total.

Note: AFT normative data for the hand-release push-up are presented in [table 14.18](#) near the end of the chapter.

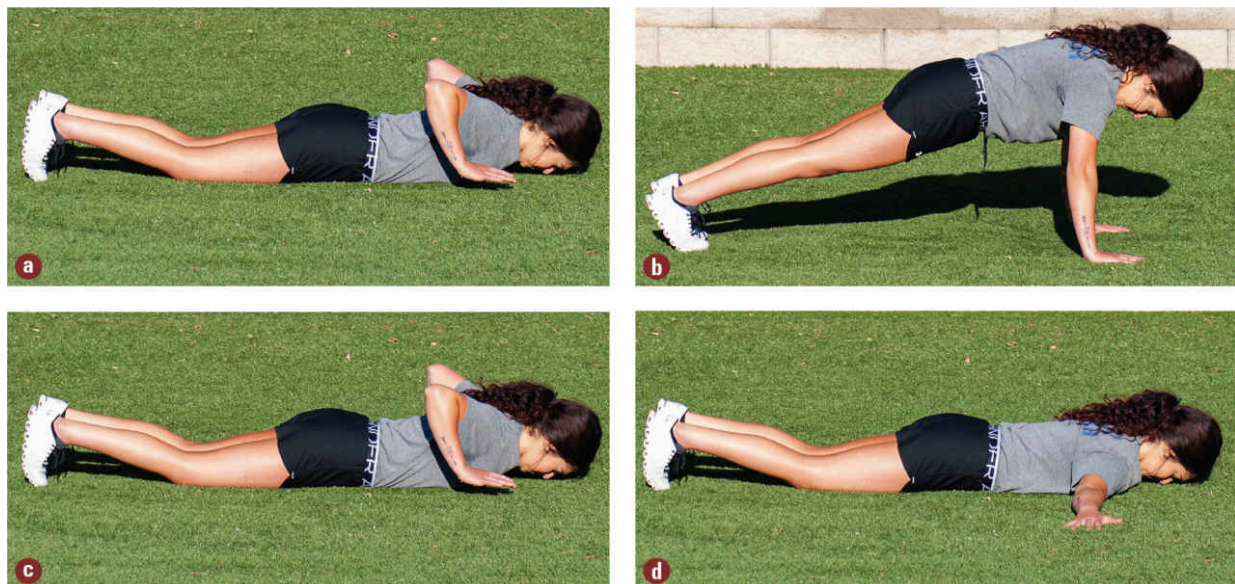


FIGURE 14.18 (a) Beginning position (but with the hands on the ground instead), (b) up position, (c) hand lift variant, and (d) arm extension variant of the hand-release push-up.

19 YMCA BENCH PRESS TEST

Equipment

- A barbell, weight plates, two safety locks, and enough total weight to assemble an 80-pound (36 kg) or a 35-pound (16 kg) load (including safety locks)
- Flat bench press bench (preferably with an upright rack to hold the barbell)
- Metronome

Personnel

- One spotter/recorder

Procedure

1. Instruct the athlete in proper technique for the flat barbell bench press as described in [chapter 16](#).
2. Allow the athlete to perform a specific warm-up of 5 to 10 repetitions with a light to moderate load.
3. The spotter/recorder stands at the head end of the bench throughout the test to help raise the bar on a failed attempt and to help the athlete place the bar back on the rack.
4. Set the resistance at 80 pounds (36 kg) for males and 35 pounds (16 kg) for females.
5. Set the metronome cadence at 60 beats/min to establish a rate of 30 repetitions per minute (one beat up, one beat down).
6. The athlete grips the bar at shoulder-width, lifts the bar off the rack, and extends the elbows. Then, in time with the metronome, the athlete repeatedly lowers the bar to the chest and raises it up again to full elbow extension, until the athlete can no longer keep up with the metronome. The movement should be smooth and controlled, with the bar reaching its highest and lowest position with each beat of the metronome.

Note: Normative data for the YMCA bench press test are presented in [table 14.19](#) near the end of the chapter.

Aerobic Capacity

20 1.5-MILE (2.4 KM) RUN

Equipment

- Stopwatch
- Quarter-mile running track or measured and marked 1.5-mile (2.4 km) flat course with a good running surface

Personnel

- One tester to call off each athlete's time, one recorder

Procedure

1. Have each athlete warm up and stretch before the test.
2. Each athlete should be recognizable to the scorer at the finish line. If that is not possible, numbers should be pinned to the athletes' shirts.
3. At the start, all athletes line up behind the starting line.
4. Instruct the athletes to complete the run as quickly as possible at a steady pace that they can barely maintain over the distance. (*Note:* Some athletes may have limited experience at pacing)

long efforts such as this, so some familiarization and prior pacing efforts in training are suggested.)

5. On an auditory signal, the athletes start running and cover the course as quickly as possible.
6. As the athletes cross the finish line, each person's time is recorded on a form as a timer calls off the time in minutes and seconds (00:00).

Note: Normative data for the 1.5-mile (2.4 km) run are presented in [tables 14.22](#) through [14.25](#) near the end of the chapter. For each 1.5-mile (2.4 km) run time, the tables show an estimated maximal rate of oxygen consumption; the norms for athletes in various sports are shown in [table 14.26](#) near the end of the chapter.

21 12-MINUTE RUN

Equipment

- A 400-meter (437-yard) track or flat looped course with a marker at each 100 meters
- Stopwatch

Personnel

- One tester to call out each athlete's position, one recorder

Procedure

1. Athletes line up at the starting line.
2. On an auditory signal, the athletes travel by foot as far as possible in 12 minutes, preferably by running but, if necessary, by walking part or all of the time.
3. At 12 minutes, on an auditory signal, all the athletes stop in place.
4. The distance run by each athlete (laps $\times$ 400 m—e.g., 5.25 laps $\times$ 400 m = 2,100 m) is calculated and recorded.

Note: Normative data for the 12-minute run are presented in [table 14.27](#) near the end of the chapter.

22 YO-YO INTERMITTENT RECOVERY TEST

The use of the Yo-Yo intermittent recovery tests (IRT1 and IRT2) is now commonplace in field testing protocols for team sports such as soccer and lacrosse (10, 18, 85). It is suggested that these tests are more specific to team sports because they mimic the demands of short, intense bursts of exercise followed by short recovery periods. Both tests consist of 2 $\times$ 20-meter (22 yd) shuttle runs at increasing speeds interspersed with a 10-second period of recovery, with the IRT1 starting at 10 km/h (6.2 mph) and the IRT2 starting at 13 km/h (8.1 mph). It is recommended that strength and conditioning professionals use the IRT1.

Equipment

- Cones
- A tape measure at least 30 meters (33 yd) long
- Audio software specifically for the Yo-Yo intermittent recovery test, IRT1 (available from a variety of commercial sources)
- Method of broadcasting the audio files (e.g., wireless speakers)
- Recording sheet
- Flat floor with good traction

Personnel

- One tester/recorder, one spotter

Procedure

1. Measure out a 20-meter (22 yd) test course and arrange cones as seen in [figure 14.19](#). Place markers 2 meters (7 ft) apart at both ends of the test course at the start and turning lines. Also measure out a 5-meter (16 ft) distance behind the start line.
2. Have the athletes warm up and stretch before the test. The athletes should run the course with a submaximal effort for practice.
3. The test begins with the athletes standing at the start line.
4. On an auditory signal, the athletes run forward to the turning line. At the sound of the second signal, athletes arrive at the turning line and then run back to the starting line, arriving in time with the next sound.
5. When the start marker is passed, the athletes jog toward the 5-meter (16 ft) mark, then turn back to the start line. At this point the athletes stop and wait for the next sound.
6. The athletes are required to place one foot on or over the starting or turning line at the sound of each beep.
7. The athletes continue running for as long as they can maintain the increasing speed as indicated by the auditory signals.
8. The termination of the test is indicated by the inability of an athlete to maintain the required pace for two trials. A warning is given the first time the start or turning line is not reached.
9. At the end of the test, record the last level and number of 2×20 -meter (22 yd) intervals performed at that level on a recording sheet.
10. The final Yo-Yo intermittent recovery speed and interval score can be used to calculate the total distance covered by the athlete during the test.

Note: Descriptive data for the Yo-Yo intermittent recovery test are presented in [table 14.28](#) near the end of the chapter.

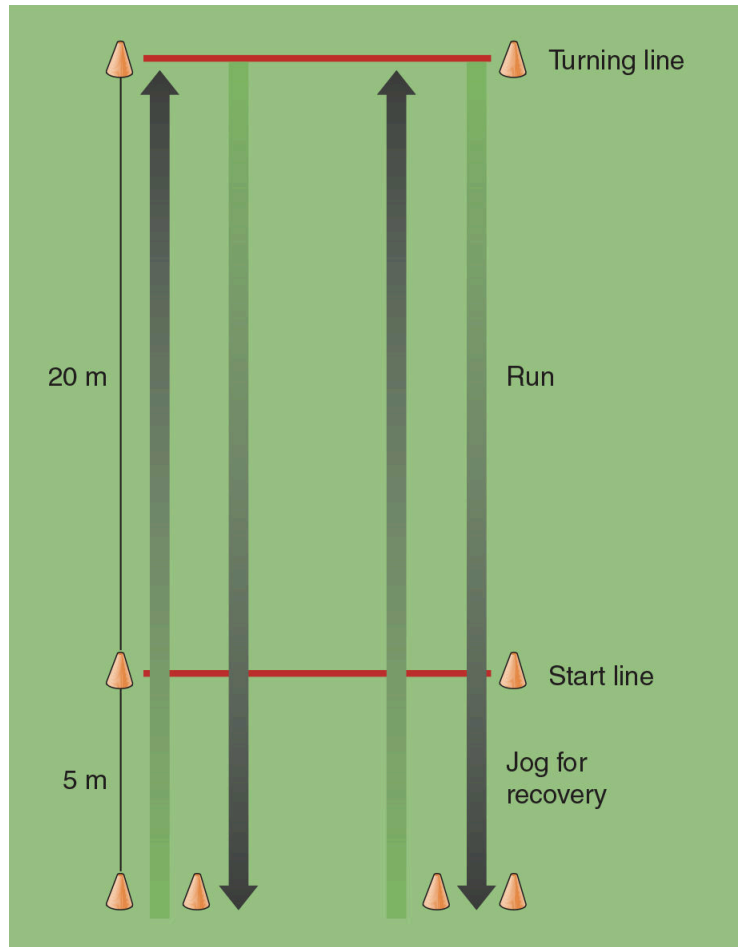


FIGURE 14.19 Setup for the Yo-Yo intermittent recovery test.

23 MAXIMAL AEROBIC SPEED TEST

Equipment

- Cones
- A tape measure at least 30 meters (33 yd) long
- Audio software specifically for the test
- Method of broadcasting the audio files (e.g., wireless speakers)
- Recording sheet
- Indoor or outdoor running track (at least 200 m [219 yd])

Personnel

- One tester/recorder

Procedure

1. Marker cones are placed at 25-meter (27 yd) intervals around the running track.
2. The initial speed of the test is set between 8 and 12 km/h (5-7 mph) depending on the fitness level of the athlete. It is generally recommended that athletes start at 10 km/h (6 mph).
3. The speed is then increased by 1 km/h (0.6 mph) every 2 minutes until the athlete cannot maintain the speed.

4. The last speed maintained for at least 2 minutes is considered the speed associated with $\dot{V}O_2\text{max}$ or MAS.
5. The test is terminated if the athlete fails to reach the next cone on two consecutive occasions in the required time.
6. The speed at the last completed stage is increased by 0.5 km/h if the athlete is able to run a half stage.
7. The $\dot{V}O_2\text{max}$ of the athlete can be calculated by multiplying $3.5 \times \text{MAS}$ (speed in km/h) (87, 88).
8. If the tester does not have access to the audio version, it is possible to conduct the test using a whistle. Calculate the timing of whistles using a set speed for reaching the next cone. For example, when the distance between cones is 25 meters, the timing of whistles for 10 km/h would be every 9 seconds.

Note: Norms for the $\dot{V}O_2\text{max}$ of athletes in various sports are shown in [table 14.26](#) near the end of the chapter.

24 30-15 INTERMITTENT FITNESS TEST (IFT)

Equipment

- A 40-meter (44 yd) long surface that is stable and suitable for an athlete to run on
- Measuring tape
- Marking cones
- 30-15 IFT audio recording and a player suitable for the media type being used that can be heard across the testing environment

Personnel

- One recorder/tester

Procedure

1. Measure out and arrange the cones as shown in [figure 14.20](#) (at lines A, B, and C).
2. Have the athletes warm up before the test by running the course with a submaximal effort for practice.
3. The test begins with the athletes standing at the start line (line A).
4. The first interval begins at a jogging speed of 8.0 km/h (5 mph). The speed is then increased by 0.5 km/h (0.3 mph) in each subsequent stage.
5. Athletes run using the beeps played from the audio file to maintain their pace. They run between lines A and C, and they should pass through each 3-meter (10 ft) zone associated with lines A, B, and C with the sound of each beep.
6. A different type of beep indicates when to stop running. At that point the athletes have 15 seconds as a recovery to walk to the next line, where they will wait for the start of the next interval.
7. Athletes advance stages until they cannot reach the relevant line on three consecutive beeps.
8. Record the running speed (km/h) achieved during the last completed stage (V_{IFT}). The total running distance of each interval is corrected by a factor of 0.7 second per change of direction.

Note: Descriptive data for the 30-15 IFT are presented in [figure 14.34](#) (*a*, females; *b*, males) near the end of the chapter.

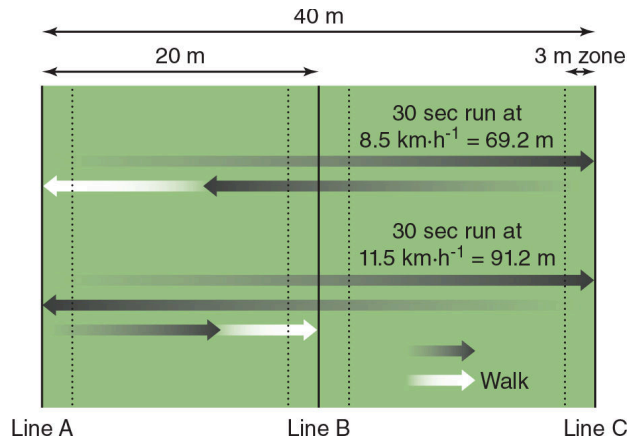


FIGURE 14.20 30-15 IFT course layout.

Reprinted by permission from M. Buchheit, "The 30-15 Intermittent Fitness Test: Accuracy for Individualizing Interval Training of Young Intermittent Sport Players," *Journal of Strength and Conditioning Research* 22, no. 2 (2008): 365-374.

Change of Direction or Agility

25 T-TEST

Equipment

- Four cones
- A tape measure at least 5 yards (4.6 m) long
- Stopwatch or timing gates, if available, with corresponding software on a laptop, tablet, or smartphone application
- Flat floor with good traction

Personnel

- One tester/recorder, one spotter

Procedure

1. Arrange four cones as shown in [figure 14.21](#) (points A, B, C, and D). Set the timing gates up at the start and finish line.
2. Have the athlete warm up and stretch before the test. The athlete may run the course with a submaximal effort for practice.
3. The test begins with the athlete standing at point A.
4. On an auditory signal, the athlete sprints forward to point B and touches the base of the cone with the right hand.
5. Then, while facing forward and not crossing the feet, the athlete shuffles to the left 5 yards (4.6 m) and touches the base of the cone at point C with the left hand.
6. The athlete then shuffles to the right 10 yards (9 m) and touches the base of the cone at point D with the right hand.
7. The athlete then shuffles to the left 5 yards (4.6 m) and touches the base of the cone at point B with the left hand, and next runs backward past point A, at which time the watch is stopped.
8. For safety, a spotter and gym mat should be positioned several feet behind point A to catch an athlete who falls while running backward.
9. The best time of two trials is recorded to the nearest 0.1 second.

10. The following are reasons for disqualification of a trial: The athlete fails to touch the base of any cone, crosses one foot in front of the other instead of shuffling the feet, or fails to face forward for the entire test.

Note: Descriptive data for the T-test are presented in [table 14.29](#) near the end of the chapter.

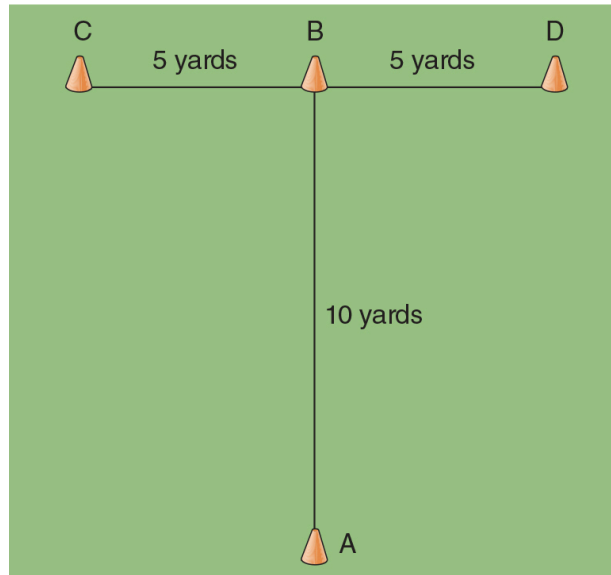


FIGURE 14.21 Floor layout for the T-test.

Adapted by permission from D. Semenic, "Tests and Measurements: The T-Test," *National Strength and Conditioning Association Journal* 12, no. 1 (1990): 36-37.

26 HEXAGON TEST

Equipment

- Adhesive tape of a color that contrasts with the floor
- Measuring tape or stick
- Stopwatch
- Flat floor with good traction

Personnel

- One timer/recorder, one line judge

Procedure

1. Using the adhesive tape, create a hexagon on the floor with 24-inch (61 cm) sides meeting to form 120-degree angles ([figure 14.22](#)).
2. The athlete warms up and practices performance of the test at submaximal speed.
3. The test begins with the athlete standing in the middle of the hexagon.
4. On an auditory signal, the athlete begins double-leg hopping from the center of the hexagon over each side and back to the center, starting with the side directly in front of the athlete, in a continuous clockwise sequence until all six sides are covered three times (three revolutions around the hexagon for a total of 18 jumps) and the athlete is again standing at the center. The athlete remains facing the same direction throughout the test.
5. If the athlete lands on a side of the hexagon rather than over it, or loses balance and takes an extra step or changes the direction in which he or she is facing, the trial is stopped and restarted after the athlete is allowed time for full recovery.

6. The best time of three trials is recorded to the nearest 0.1 second.

Note: Descriptive data for the hexagon test are presented in [table 14.29](#) near the end of the chapter.

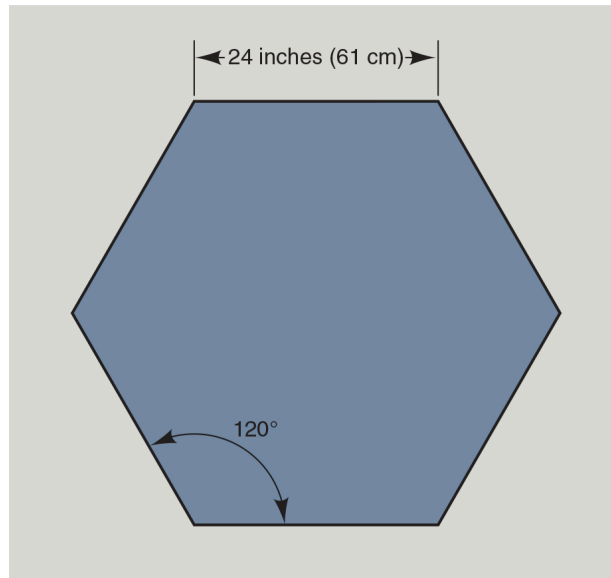


FIGURE 14.22 Layout for the hexagon test.

Adapted by permission from K. Pauole, K. Madole, J. Garhammer, et al., "Reliability and Validity of the T-Test as a Measure of Agility, Leg Power, and Leg Speed in College-Aged Men and Women," *Journal of Strength and Conditioning Research* 14, no. 4 (2000): 443-450.

27 5-10-5 TEST

This test is also called the *pro agility test* or the *20-yard shuttle*.

Equipment

- An American football field or other field marked with three parallel lines 5 yards (4.6 m) apart ([figure 14.23](#))
- Stopwatch or timing gates, if available, with corresponding software on a laptop, tablet, or smartphone application

Personnel

- One timer/recorder, one line judge

Procedure

1. If using one timing gate for total end time, set this up at the start and finish line. If multiple gates are available, set up at the start and finish line as well as at each turn line to measure each split.
2. The athlete straddles the center line using a three-point stance.
3. On an auditory signal, the athlete sprints 5 yards (4.6 m) to the line on the left, then changes direction and sprints 10 yards (9 m) to the line on the right, then again changes direction and sprints 5 yards (4.6 m) to the center line. Hand (or foot) contact must be made with all indicated lines. (*Note:* It is important that this is kept consistent for both trials.)
4. The best time of two trials is recorded to the nearest 0.01 second.

Note: Normative data for the 5-10-5 test (previously pro agility) are presented in [tables 14.29](#) and [14.30](#) near the end of the chapter.

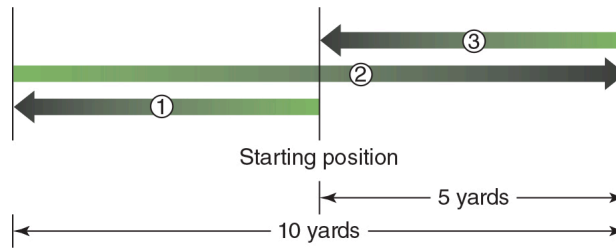


FIGURE 14.23 Layout for the 5-10-5 test.

28 5-0-5 TEST

Equipment

- Seven cones
- Stopwatch or timing gates, if available, with corresponding software on a laptop, tablet, or smartphone application

Personnel

- One timer/recorder, one line judge

Procedure

1. Arrange the cones as seen in [figure 14.24](#). If timing lights are available, these can also be set up as shown.
2. Have the athlete warm up and stretch before the test. The athlete may run the course with a submaximal effort for practice.
3. The test begins with the athlete standing at the start line.
4. On an auditory signal, the athlete sprints forward 10 meters (11 yd) to the first set of timing lights, then sprints a further 5 meters (5 yd) to the turning line (one foot must be on or over the line), where he or she is required to turn and accelerate off the line.
5. The athlete may slow down only after passing through the timing lights for the second time.
6. The best time of two trials is recorded to the nearest 0.1 second.
7. The athlete completes the trials turning off the preferred leg. Alternatively, trials (at least two) can be administered turning off either the preferred or non-preferred leg.

Note: Descriptive data for the 5-0-5 test is presented in [tables 14.29](#) and [14.30](#) near the end of the chapter.

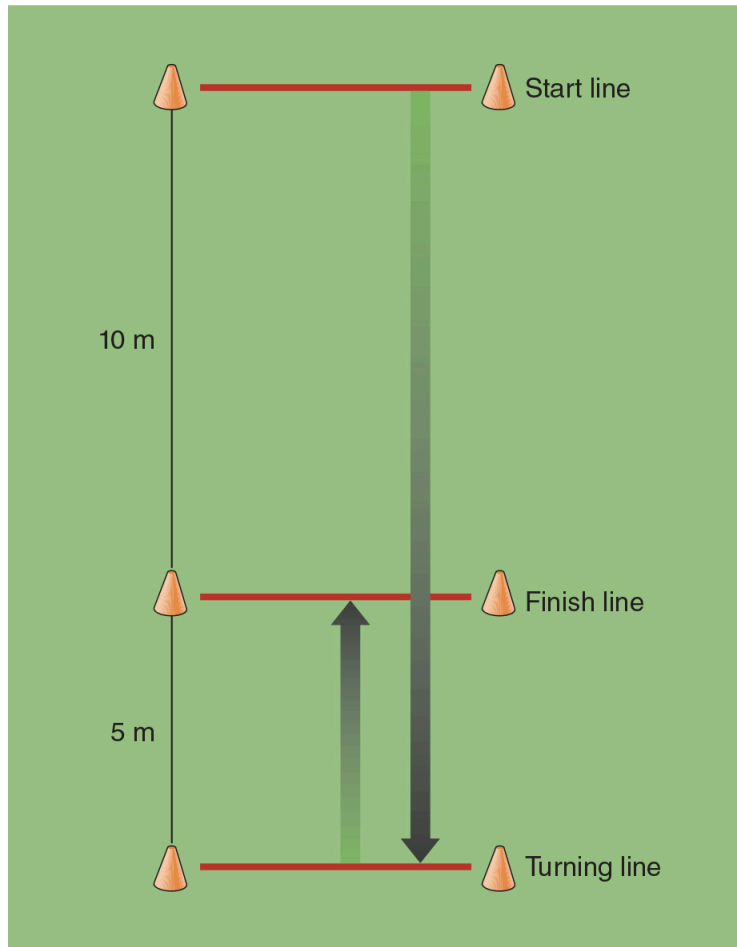


FIGURE 14.24 Layout for the 5-0-5 test.

29 ILLINOIS TEST

Equipment

- Stopwatch or timing lights
- Tape measure
- Eight cones

Personnel

- One tester/recorder

Procedure

1. Set up the cones on a course of 10 meters (11 yd) in length by 5 meters (5 yd) in width. Four cones should be on the corners of the 10-meter $\times$ 5-meter square, with the additional four placed in the center of the 5-meter width equally spaced along the 10-meter length. This will create three aisles of cones in the order of two, four, and two through which the athlete will run ([figure 14.25](#)).
2. Instruct the athlete to lie prone at the cone to the outside left of the full 10-meter length. The athlete will be instructed to get up and run when the tester says "Go," sprint to the cone 10 meters ahead, and then turn around to sprint back to the cones in the middle aisle.